UCCM1153

Introduction to Calculus and Applications

Chapter 1

Functions and Models

Contents

1.1 Functions

- Definitions
- Representations of Functions
- Piecewise Defined Functions
- Even and Odd Functions
- Increasing and Decreasing Functions

Goals

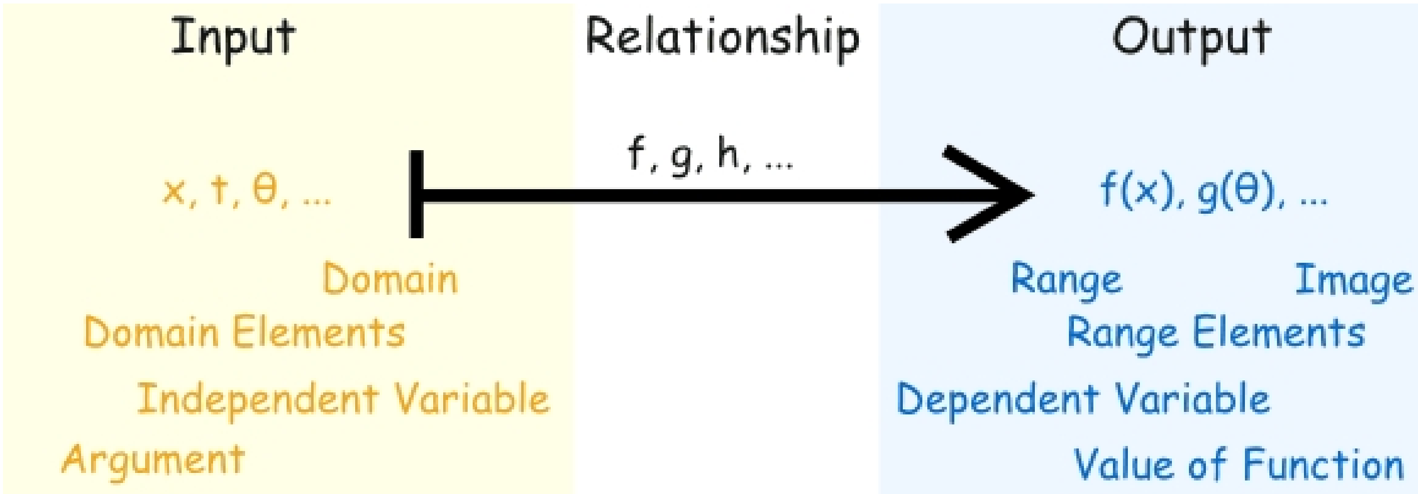
- Learn to represent functions using

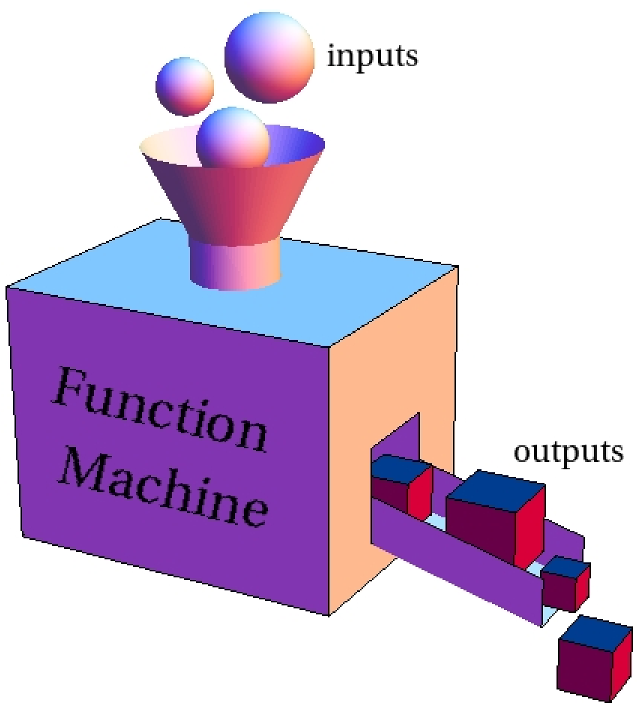
  Words, Tables of values, Graphs and Formulas

- Discuss piecewise defined functions, including the absolute value function

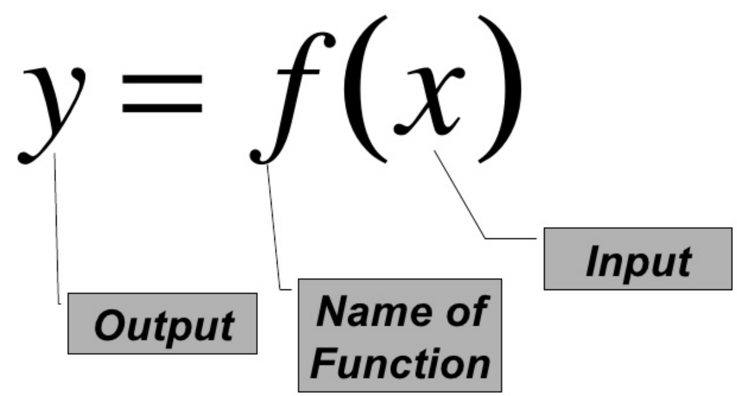
- Study symmetry of functions, including even and odd functions

- Discuss increasing and decreasing functions





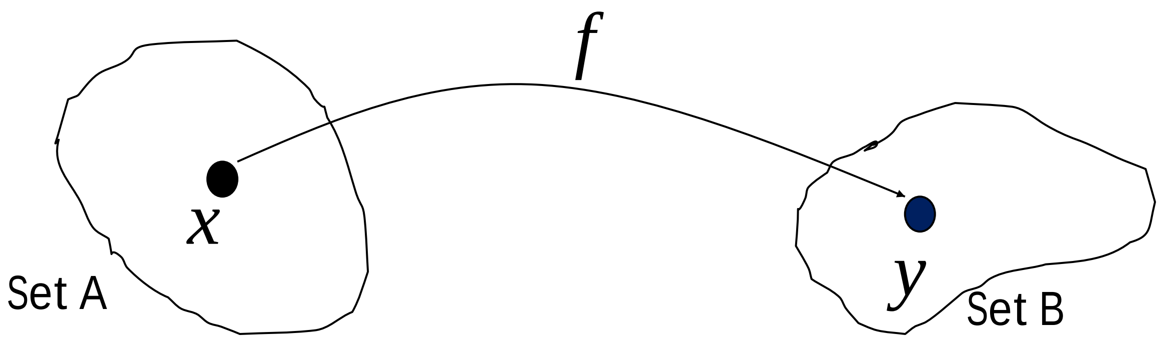
Function Notation



Functions

- A function is a rule that assigns *exactly* one element y in a set B to each element x in a set A.

$$x \longrightarrow \text{Function} \longrightarrow y$$



Notation: $\boxed{y = f(x)}$

- ➢ x is an independent variable
- ➢ y is an dependent variable.

E.g.:The function which square a number and adds on 5, can be written as $f(x) = x^2 + 5$

Functions(Cont')

The phrase "y is a function of x" means that the value of y depends upon the value of x, so:

- y can be written in terms of x (e.g. $y = 3x$).
- If $f(x) = 3x$, and y is a function of x (i.e. $y = f(x)$), then the value of y when x is 4 is $f(4)$, which is found by replacing x"s by 4"s .
- One value of x is assigned to exactly one value of y.

➢ E.g.: If $f(x) = 3x + 4$, find $f(5)$ and $f(x + 1)$.

$f(5) = 3(5) + 4 = 19$

$f(x + 1) = 3(x + 1) + 4 = 3x + 3 + 4 = 3x + 7$

Domain and Range for a function $y = f(x)$

- **Domain** is the possible values can be taken by independent variable, x.

- **Range** of f is the set of all possible values of $f(x)$ as x varies throughout the domain.

- The most common method for visualizing a function is its graph.

- If f is a function with domain A, then its graph is the set of ordered pairs $\{(x, f(x)) | x \in A\}$

Example: "Multiply by 2" is a very simple function.

Here are the three parts:

Input	*Relationship*	Output
0	× 2	0
1	× 2	2
7	× 2	14
10	× 2	20
...	...	...

For an input of 50, what is the output?

Example: with $\mathbf{f(x) = x^2}$:

- an input of 4
- becomes an output of 16.

In fact we can write $\mathbf{f(4) = 16}$.

Graph of a Function

- The *graph* of a function f is the set of ordered pairs $\{(x, f(x)) \mid x \in A\}$
- So the graph consists of all points (x, y) where $y = f(x)$ and x is in the domain of f.
- From the graph of f we can read…
  - the value of $f(x)$ as being the height of the graph above the point x;
  - the domain and range of f,

Graph of a Function (Cont')

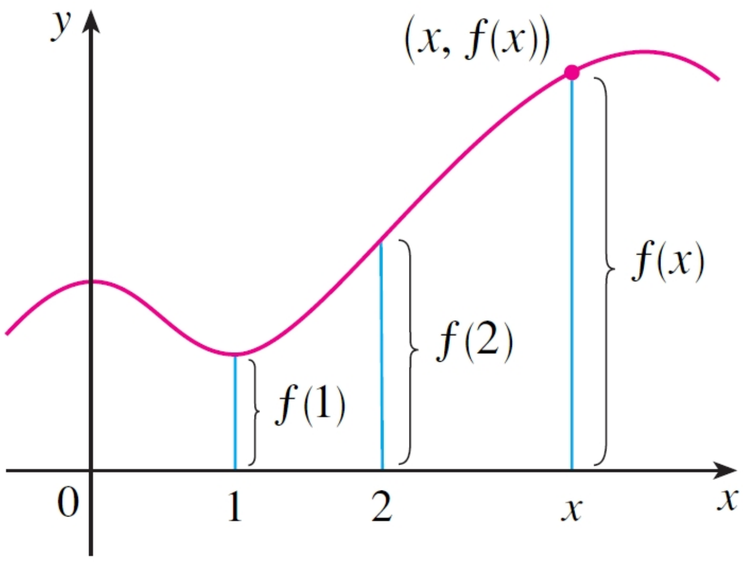


FIGURE 4

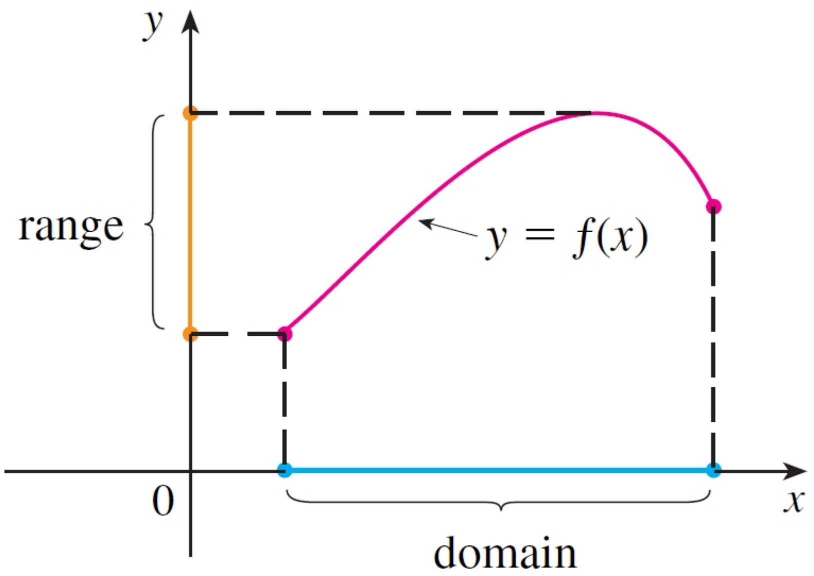


FIGURE 5

Representations of Functions

- Functions can be represented in four ways:
  - Verbally (that is, by a description in words)
  - Numerically (by a table of values)
  - Visually (by a graph)
  - Algebraically (by an explicit formula)
- It is often useful to convert from one representation to another, where possible.

Representations of Functions (Cont')

- The world population as a function of time was given verbally.
- The area of a circle as a function of its radius (r) was given algebraically, i.e. $A = \pi r^2$
- The cost of mailing a letter as a function of weight was also given numerically in a table shown :

w (ounces)	$C(w)$ (dollars)
$0 < w \leq 1$	0.37
$1 < w \leq 2$	0.60
$2 < w \leq 3$	0.83
$3 < w \leq 4$	1.06
$4 < w \leq 5$	1.29
$\vdots$	$\vdots$

Representations (cont'd)

– The vertical acceleration of the ground in an earthquake, as a function of time, was given <u>visually</u>.

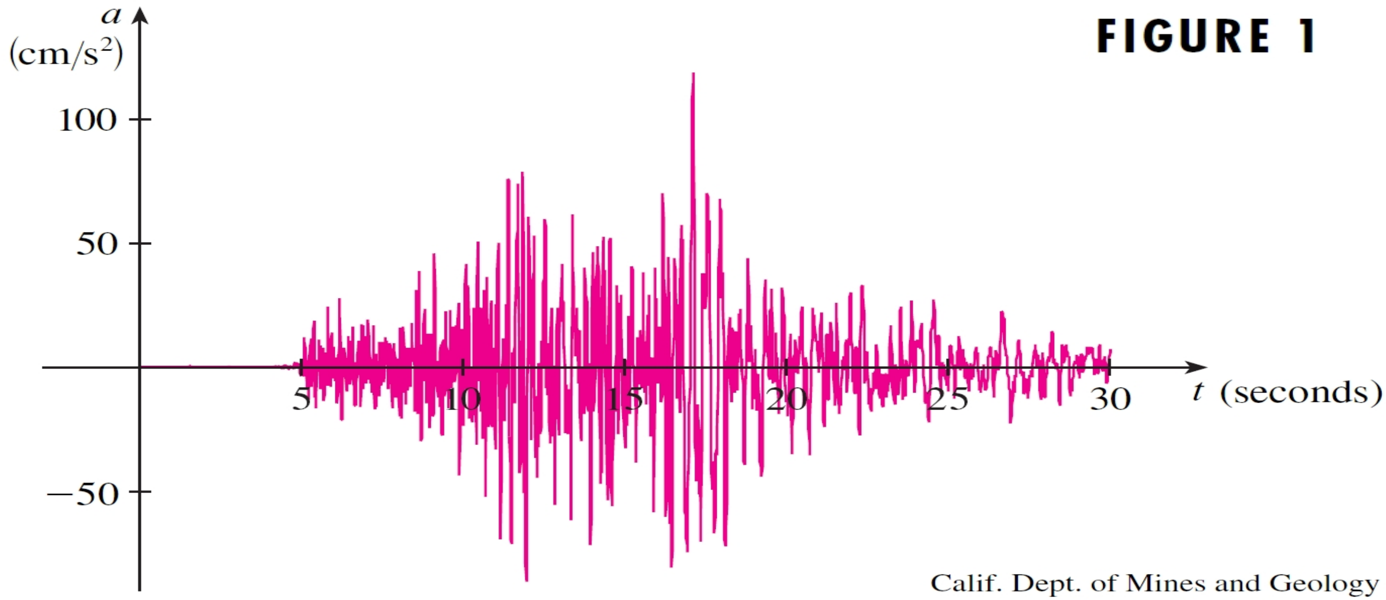


FIGURE 1

Calif. Dept. of Mines and Geology

Example A

- The data shown give the concentration (in moles per liter) after t minutes of an acid used in an experiment.
  - Use the data to draw a rough graph of the concentration function, and then...
  - ...use this graph to estimate the concentration after 5 minutes.

t	$C(t)$
0	0.0800
2	0.0570
4	0.0408
6	0.0295
8	0.0210

Example A (Cont')

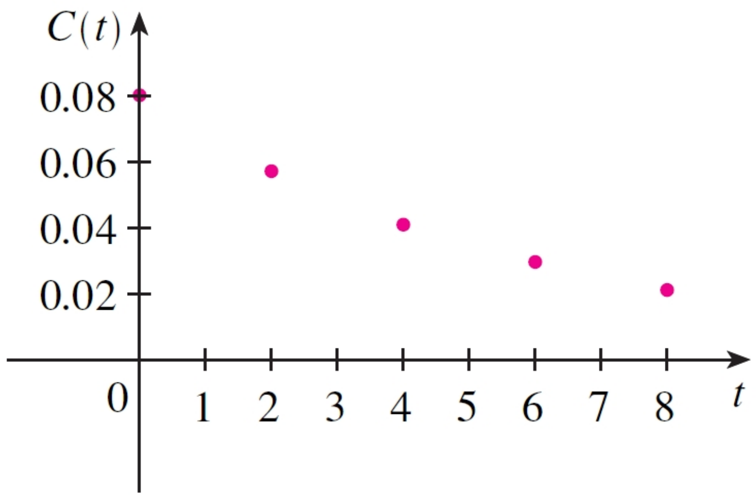


FIGURE 14

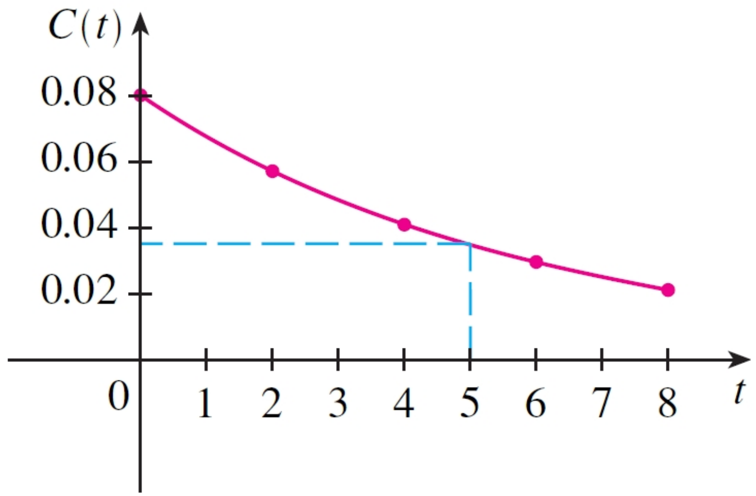


FIGURE 15

Solution

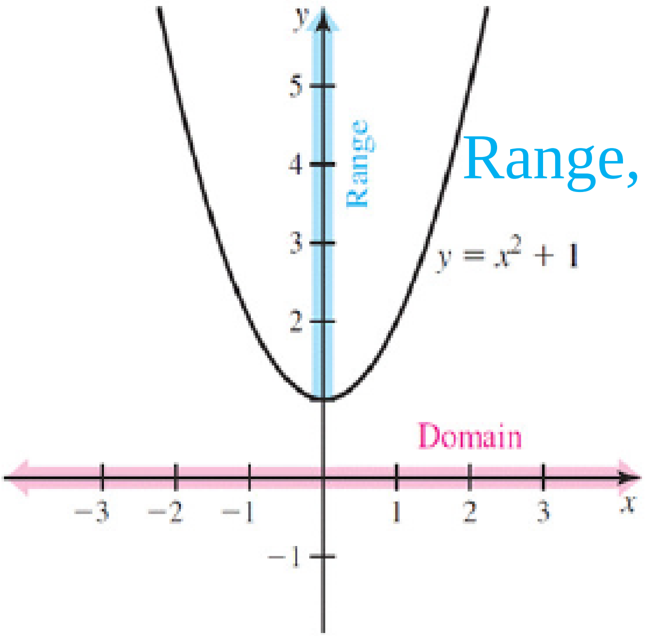
- Then we use the graph to estimate that the concentration after 5 minutes is

$$C(5) \approx 0.035 \text{ mole/liter.}$$

t	$C(t)$
0	0.0800
2	0.0570
4	0.0408
6	0.0295
8	0.0210

Examples

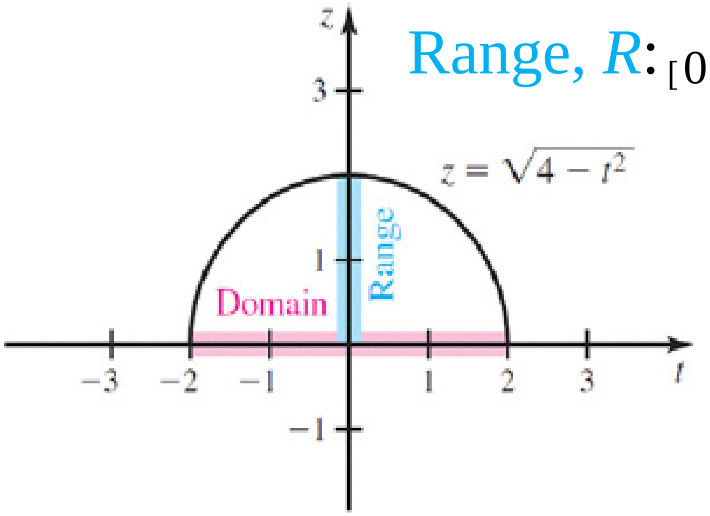
Example 1.1



Range, R: $[1, \infty)$

Domain, D: $\mathfrak{R}$

Example 1.2



Range, R: $[0, 2]$

Domain, D: $t \geq -2, t \leq 2$

Examples (Cont')

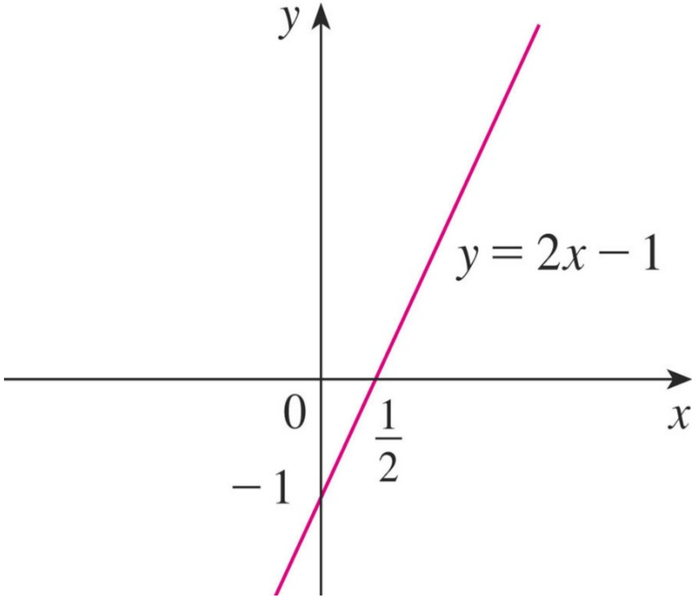
***Example* 1.3**

Sketch the graph and find the domain and range of each function.

(a) $f(x) = 2x\text{-}1$

Domain: $\mathfrak{R}$

Range: $\mathfrak{R}$

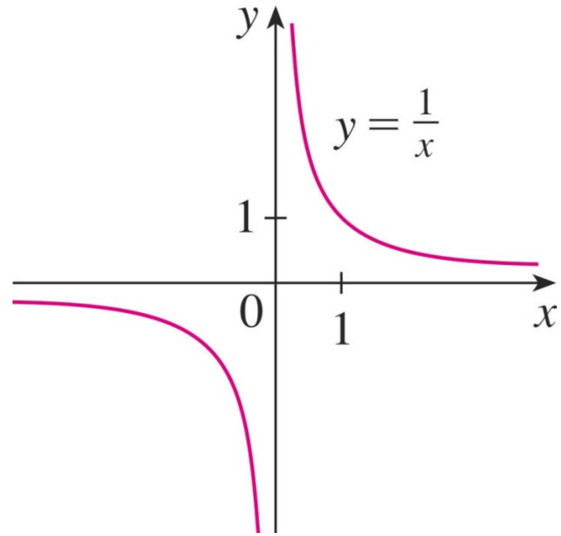


Examples (Cont')

(b) $f(x) = \dfrac{1}{x}$

Domain : $(-\infty, 0) \cup (0, \infty)$

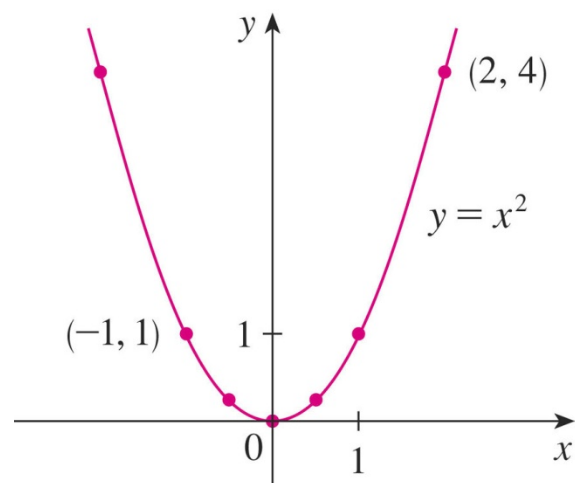
Range : $(-\infty, 0) \cup (0, \infty)$



(c) $f(x) = x^2$

Domain: $\Re$

Range: $\{y \mid y \geq 0\} = [0, \infty)$

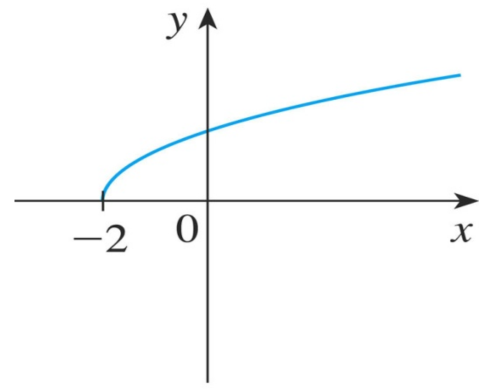


Examples (Cont')

(d) $f(x) = \sqrt{x+2}$

Domain: $[-2, \infty)$

Range: $[0, \infty)$



$y = \sqrt{x+2}$

(e) $f(x) = \dfrac{x+1}{x-1}$

Domain: $\{x : x \neq 1\} = (-\infty, 1) \cup (1, \infty)$

Range:

$$\frac{x+1}{x-1} = y$$

$$(x-1)y = x+1$$

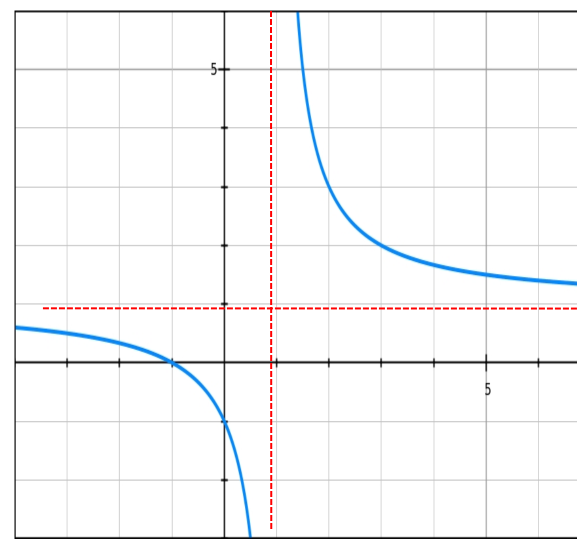
$$xy - y = x+1$$

$$xy - x = y+1$$

$$x(y-1) = y+1$$

$$x = \frac{y+1}{y-1}$$

$$\{y : y \neq 1\} = (-\infty, 1) \cup (1, +\infty)$$

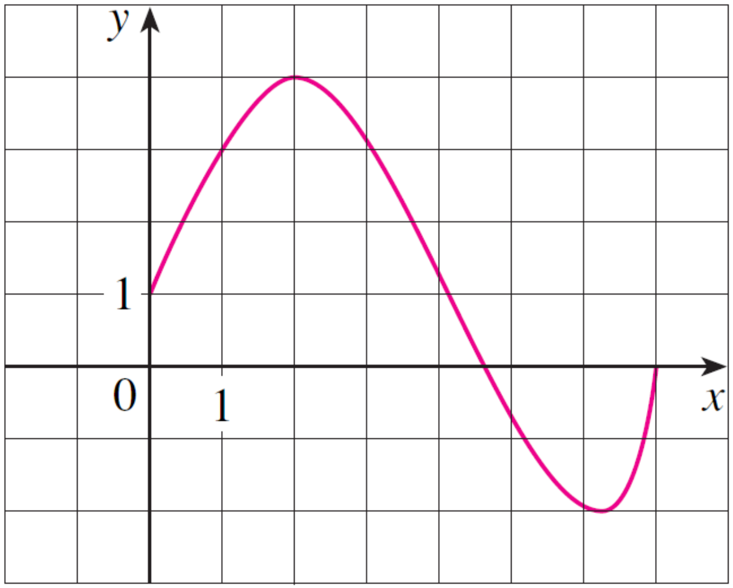


Example

- The graph of a function *f* is shown at right.

a) Find the values of $f(1)$ and $f(5)$.

b) What are the domain and range of *f* ?



Example (cont'd)

- <u>Solution</u>

a) We see from the figure that the point (1, 3) lies on the graph of f, so the value of f at 1 is $f(1) = 3$.

- Likewise when $x = 5$, the graph lies about 0.7 unit below the x-axis, so we estimate that $f(5) \approx -0.7$.

c) Since $f(x)$ is defined when $0 \le x \le 7$, the domain of f is the closed interval $[0, 7]$. Also, f takes on all values from -2 to 4, so the range of is $[-2, 4]$.

Piecewise-defined functions

- The function with different formulas in different parts of its domain.

***Example* 1.4**

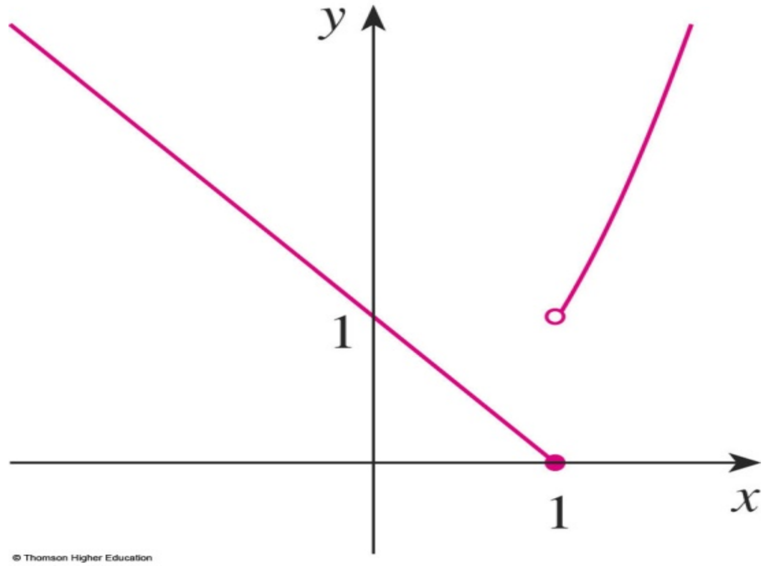
$$f(x)=\begin{cases} 1-x, & \text{if } x\le 1 \\ x^2, & \text{if } x>1 \end{cases}$$

Evaluate $f(0)$, $f(1)$, and $f(2)$ and sketch the graph.

- Since $0 \le 1$, we have $f(0) = 1 - 0 = 1$.
- Since $1 \le 1$, we have $f(1) = 1 - 1 = 0$.
- Since $2 > 1$, we have $f(2) = 2^2 = 4$.
- If $x \le 1$, then $f(x) = 1 – x$. So, the part of the graph of f that lies to the left of the vertical line $x = 1$ must coincide with the line $y = 1 - x$, which has slope -1 and y-intercept 1.

Piecewise-defined functions (Cont')

- If $x > 1$, then $f(x) = x^2$. So, the part of the graph of f that lies to the right of the line $x = 1$ must coincide with the graph of $y = x^2$, which is a parabola.
- The solid dot indicates that the point (1, 0) is included on the graph.
- The open dot indicates that the point (1, 1) is excluded from the graph.



Absolute value

- The absolute value of a number a, $|a|$, is the distance from a to 0 on the real number line. Distances are always positive or 0, so we have $|a|$ 0 for every number a.

<u>Example</u> **<u>1.5</u>**

Evaluate

(a) |3| = 3 (b) |-3| = 3 (c) |0| = 0

(d) $|\sqrt{2}-1| = \sqrt{2}-1$ (e) $|3-\pi| = \pi - 3$

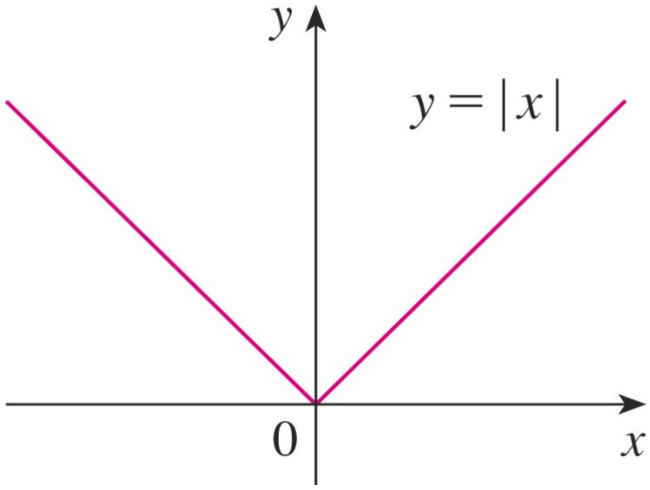
Example **1.6**

Sketch the graph of the absolute value function $f(x) = |x|$.

$$|x| = \begin{cases} x, if\, x \geq 0 \\ -x, if\, x < 0 \end{cases}$$

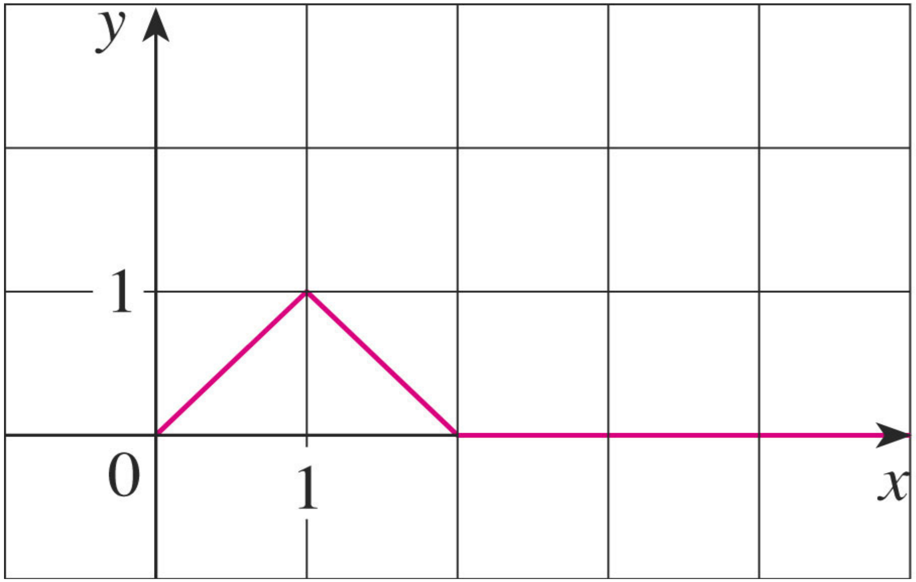
The line $y = x$ to the right of the y-axis

The line $y = -x$ to the left of the y-axis



***Example* 1.7**

Find a formula for the function *f* graphed in the figure.



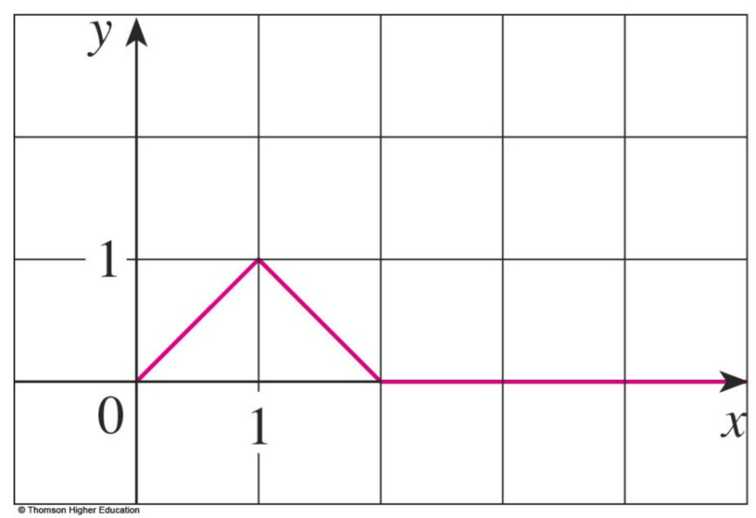
***Example* 1.7** (Cont')

The line through (0, 0) and (1, 1) has slope $m = 1$ and y-intercept $b = 0$.

So, its equation is $y = x$.

- Thus, for the part of the graph of f that joins (0, 0) to (1, 1), we have:

$$f(x) = x \quad \text{if } 0 \le x \le 1$$



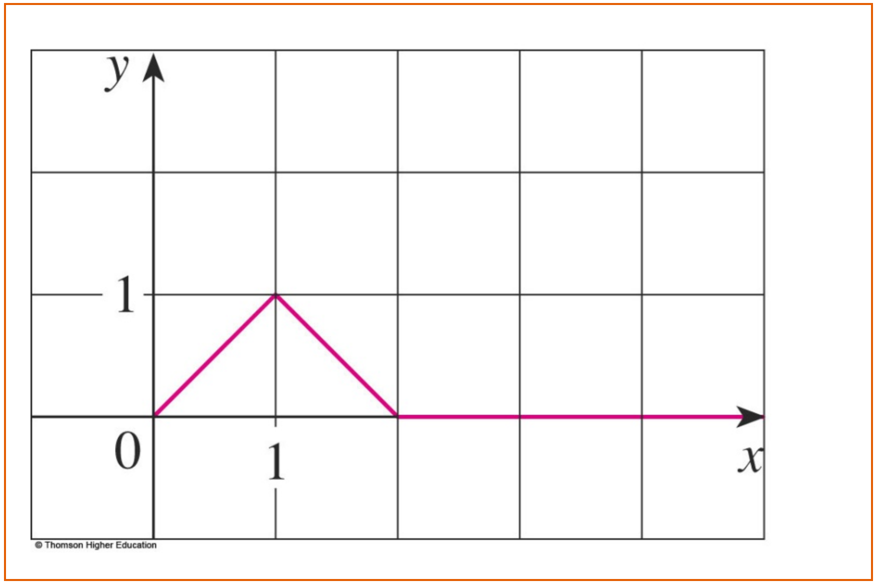
Example **1.7** (Cont')

The line through (1, 1) and (2, 0) has slope $m = -1$.

So, its point-slope form is $y - 0 = (-1)(x - 2)$ or $y = 2 - x$.

– So, we have:

$$f(x) = 2 - x \quad \text{if } 1 \le x \le 2$$

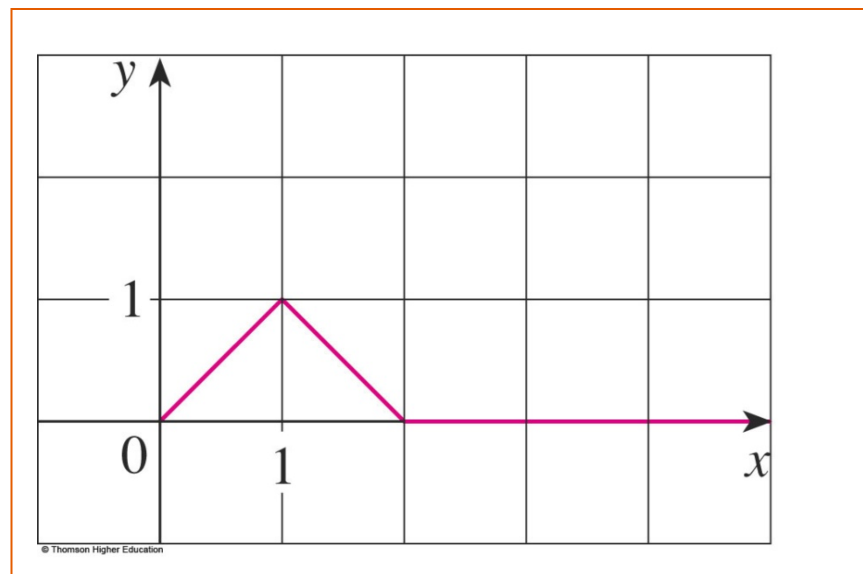


***Example* 1.7** (Cont'd)

We also see that the graph of *f* coincides with the *x*-axis for *x* > 2.

Putting this information together, we have the following three-piece formula for *f*:

$$f(x) = \begin{cases} x & \text{if } 0 \le x \le 1 \\ 2 - x & \text{if } 1 < x \le 2 \\ 0 & \text{if } x > 2 \end{cases}$$



Symmetry

Odd and even functions are special types of functions with special characteristics.

The trick to working with *odd* and *even* functions is to remember to plug in $(-x)$ in place of x and see what happens.

Examine $f(-x)$.

If there is no change = even. If there is a negation = odd.

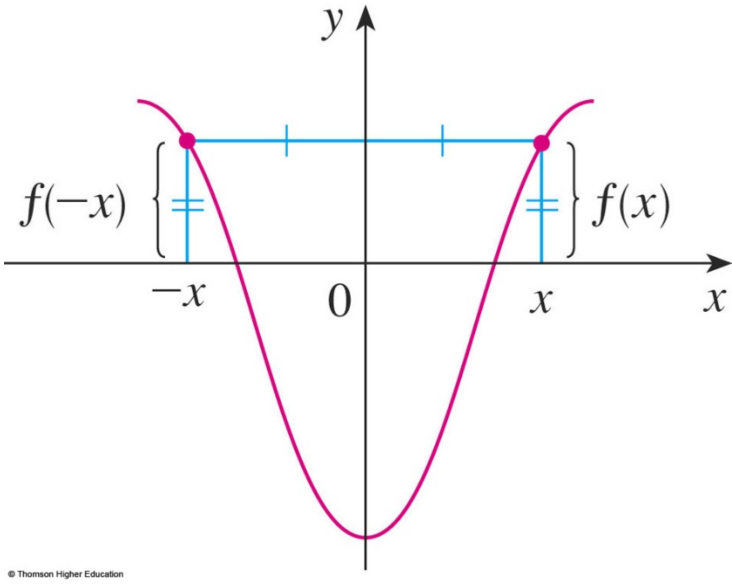
Symmetry

(1) Even function

- If $f(-x) = f(x)$, x is in the domain of $f(x)$.
- The geometric significance of an even function is that its graph is symmetric with respect to the y-axis.

$$\boxed{(x, y) \longrightarrow (-x, y)}$$

- This means that, if we have plotted the graph of f for $x \geq 0$, we obtain the entire graph simply by reflecting this portion about the y-axis.



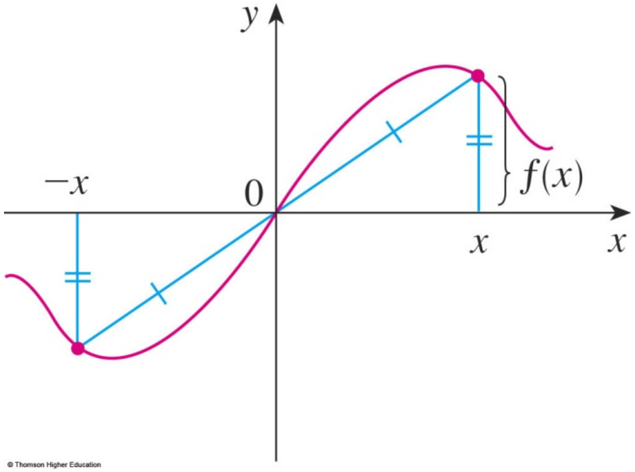
Symmetry

(2) Odd function

- If $f(-x) = -f(x)$, x is in the domain of $f(x)$.
- The graph of an odd function is symmetric about the origin (0,0).

$$\boxed{(x,y) \longrightarrow (-x,-y)}$$

- If we already have the graph of f for $x \geq 0$, we can obtain the entire graph by rotating this portion through 180° about the origin.

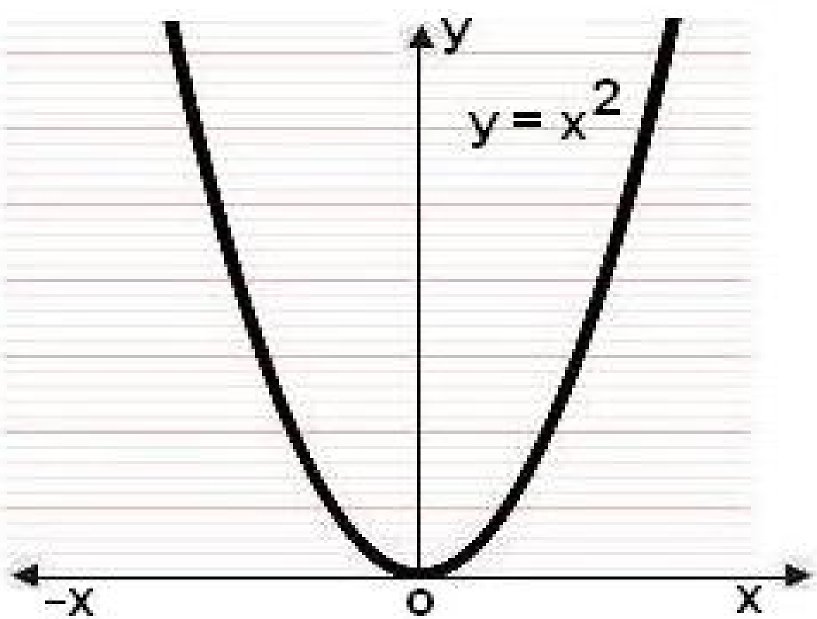


***Example* 1.8**

Determine if the following functions are odd or even.

(a) $f(x) = x^2$

$f(-x) = (-x)^2 = x^2 = f(x)$



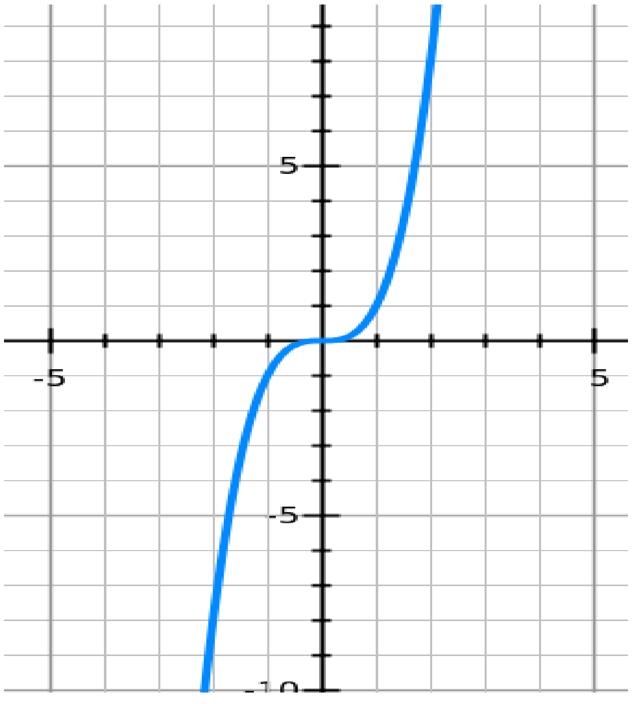
The function $f(x) = x^2$ is even.

***Example* 1.8(Cont')**

(b) $f(x) = -x^3$

The function $f(x) = x^3$ is odd because

$f(-x) = (-x)^3 = -x^3 = -f(x)$.

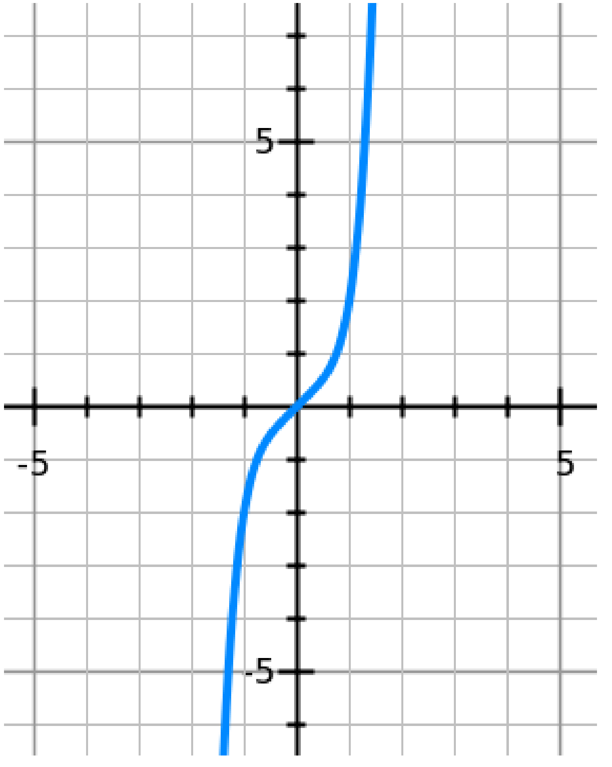


Example 1.8(Cont')

(c) $f(x) = x^5 + x$

$$f(-x) = (-x)^5 + (-x)$$
$$= (-1)^5 x^5 + (-x)$$
$$= -x^5 - x$$
$$= -(x^5 + x)$$
$$= -f(x)$$

Thus, $f(x)$ is an odd function.

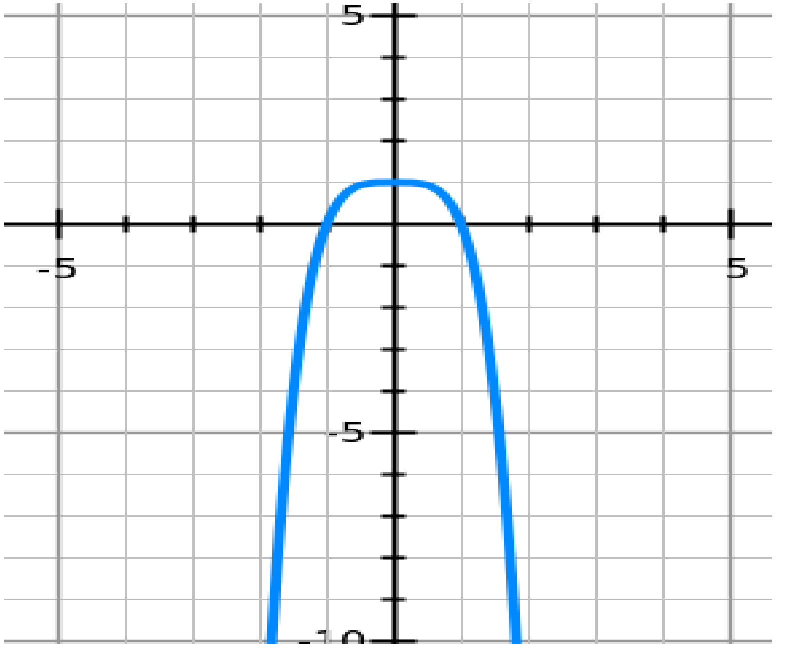


***Example* 1.8(Cont')**

(d) $g(x) = 1 - x^4$

$$g(-x) = 1 - (-x)^4 = 1 - x^4 = g(x)$$

So, $g(x)$ is even.

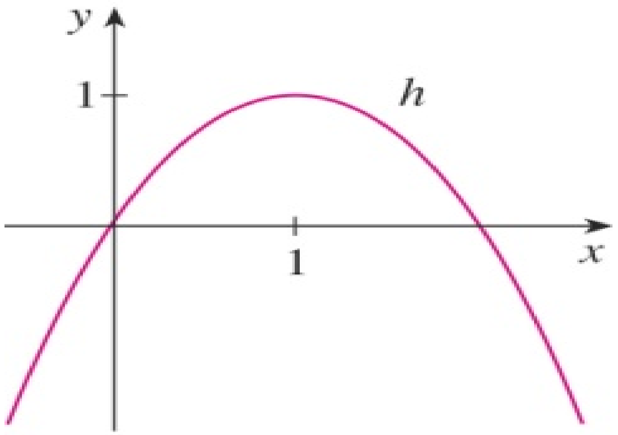


Example **1.8(Cont')**

(e) $h(x) = 2x - x^2$

$$h(-x) = 2(-x) - (-x)^2 = -2x - x^2$$

Since $h(-x) \neq h(x)$ and $h(-x) \neq -h(x)$, we conclude that h is neither even nor odd.



Example (TRY)

$$g(t) = \begin{cases} 3t^2 + 4 & \text{if } t \le -4 \\ 10 & \text{if } -4 < t \le 15 \\ 1 - 6t & \text{if } t > 15 \end{cases}$$

evaluate each of the following.

(a) $g(-6)$

(b) $g(-4)$

(c) $g(1)$

(d) $g(15)$

(e) $g(21)$

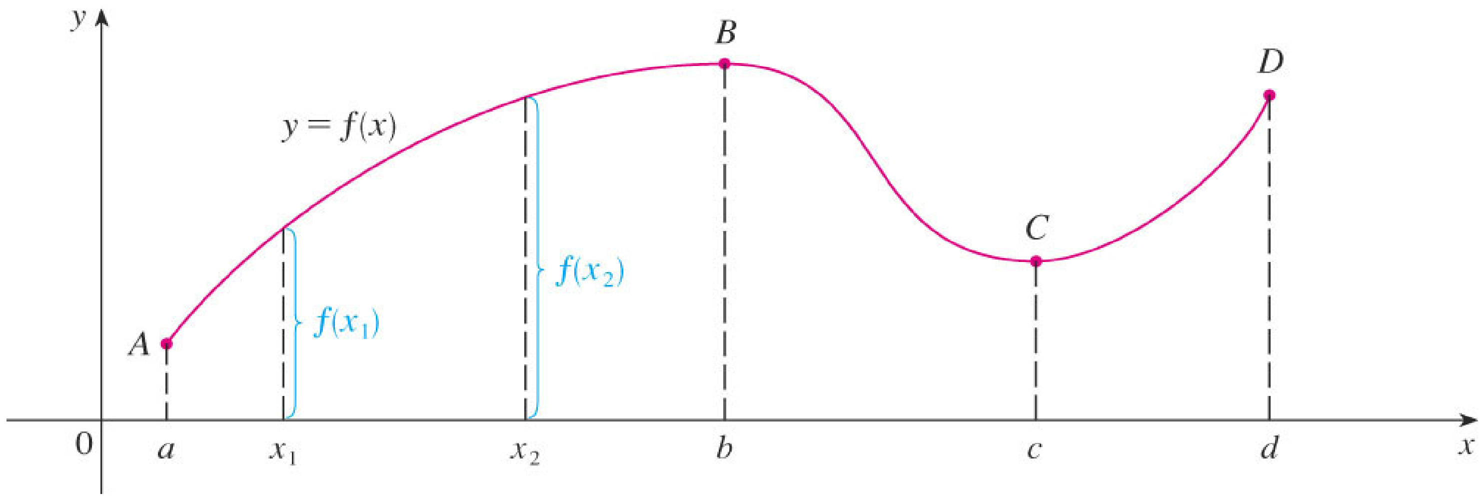
Increasing and decreasing functions

Definition:

- A function f is increasing on an interval I if $\boxed{f(x_1) < f(x_2)}$ for $\boxed{x_1 < x_2}$ in I.
  - When x increases, $f(x)$ also increases.
  - $f(x)$ is moving up at the interval I
- A function f is decreasing on the interval I if $\boxed{f(x_1) > f(x_2)}$ for $\boxed{x_1 < x_2}$ in I.
  - When x increases, $f(x)$ decreases.
  - $f(x)$ is moving down at the interval I

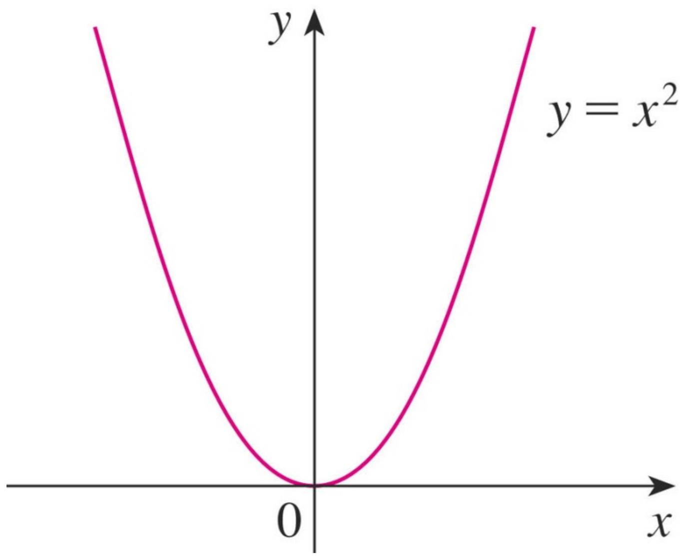
Increasing and decreasing functions (Cont')

- This graph rises from A to B, falls from B to C, and rises again from C to D.
- The function f is said to be increasing on the interval $[a, b]$, decreasing on $[b, c]$, and increasing again on $[c, d]$.
- Notice that, if x_1 and x_2 are any two numbers between a and b with $x_1 < x_2$, then $f(x_1) < f(x_2)$.
- We use this as the defining property of an increasing function.



***Example* 1.9**

$$y = x^2$$



- You can see from the figure that the function $f(x) = x^2$ is decreasing on the interval $(-\infty, 0]$ and increasing on the interval $[0, \infty)$.

Class Activity

The graphs of *f* and *g* are given.

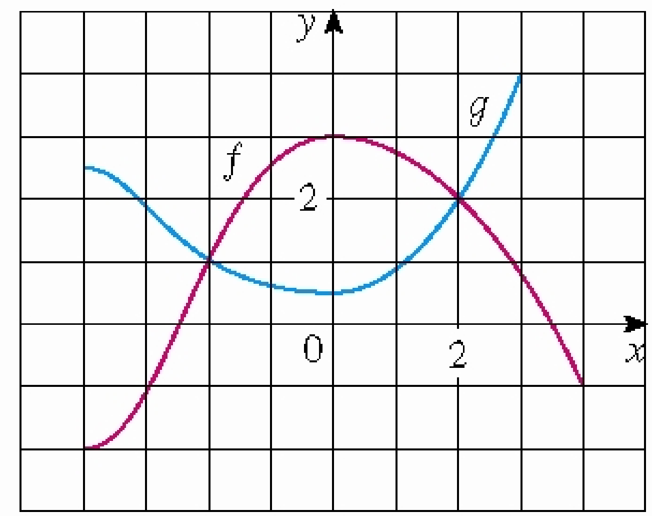
(a) State the values of $f(-4)$ and $g(3)$.

(b) For what values of x is $f(x) = g(x)$?

(c) Estimate the solution of the equation $f(x) = -1$

(d) State the domain and range of *f*.

(e) State the domain and range of *g*.



(a) The point $(-4,-2)$ is on the graph of f, so $f(-4)=-2$. The point $(3,4)$ is on the graph of g, so $g(3)=4$.

(b) We are looking for the values of x for which the y-values are equal. The y-values for f and g are equal at the points $(-2,1)$ and $(2,2)$, so the desired values of x are -2 and 2.

(c) $f(x)=-1$ is equivalent to $y=-1$. When $y=-1$, we have $x=-3$ and $x=4$.

(e) The domain of f consists of all x-values on the graph of f. For this function, the domain is $-4 \le x \le 4$, or $[-4,4]$. The range of f consists of all y-values on the graph of f. For this function, the range is $-2 \le y \le 3$, or $[-2,3]$.

(f) The domain of g is $[-4,3]$ and the range is $[0.5,4]$.

(a) The point $(-4,-2)$ is on the graph of f , so $f(-4)=-2$. The point $(3,4)$ is on the graph of g , so $g(3)=4$.
(b) We are looking for the values of x for which the y –values are equal. The y –values for f and g are equal at the points $(-2,1)$ and $(2,2)$, so the desired values of x are -2 and 2 .
(c) $f(x)=-1$ is equivalent to $y=-1$. When $y=-1$, we have $x=-3$ and $x=4$.
(d) As x increases from 0 to 4 , y decreases from 3 to -1 . Thus, f is decreasing on the interval $[0,4]$.
(e) The domain of f consists of all x –values on the graph of f . For this function, the domain is $-4 \leq x \leq 4$, or $[-4,4]$. The range of f consists of all y –values on the graph of f . For this function, the range is $-2 \leq y \leq 3$, or $[-2,3]$.
(f) The domain of g is $[-4,3]$ and the range is $[0.5,4]$.

Section 1.2

Models and curve fitting

Models and Curve Fitting

- Mathematical Model
- The types of Functions:
  - Linear
  - Polynomial
  - Power and root
  - Rational
  - Algebraic
  - Trigonometric
  - Exponential and logarithmic
  - Transcendental

Mathematical Models

- A *mathematical model* is a mathematical description of a real-world phenomenon, such as:
  - the size of a population
  - the demand for a product
  - the speed of a falling object
  - the concentrations of a product in a chemical reaction
- The purposes of the model are to…
  - understand the phenomenon;
  - make predictions about future behavior.

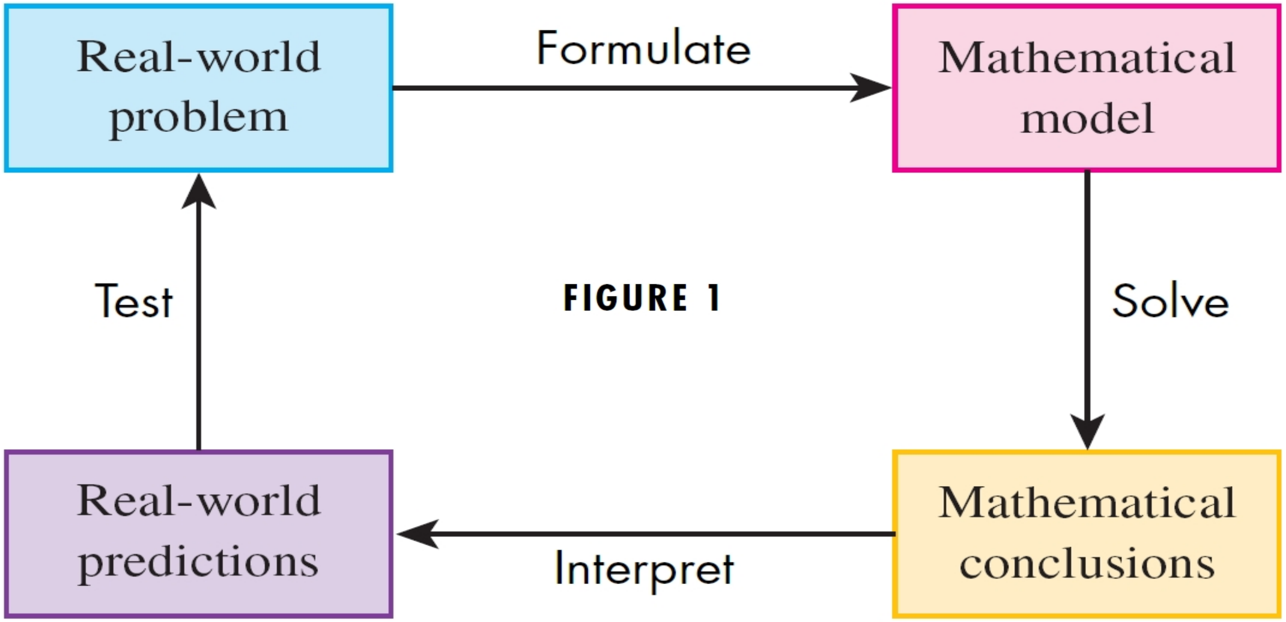
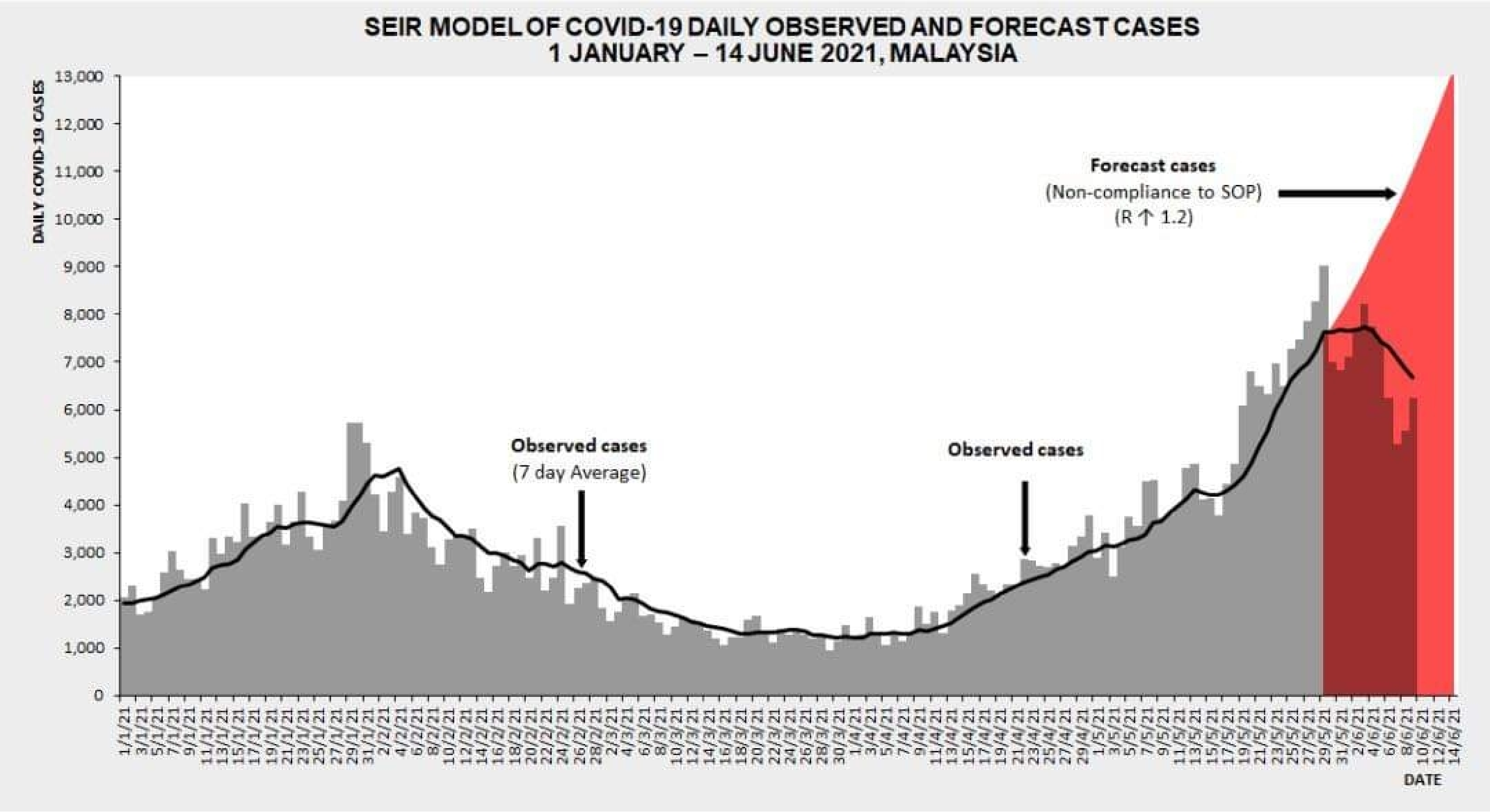


FIGURE 1

- Figure 1 illustrates the modeling process:
  - First we formulate the model;
  - Next we solve the model mathematically;
  - Then we interpret the conclusions;
  - Finally we test our predictions—
    - and repeat the cycle as needed!



A deterministic SEIR model based on the clinical progression of the disease, epidemiological status of the individuals, and intervention measures was proposed. The mathematical model used was the Susceptible, Exposed, Infectious and Removed (SEIR) model**,** as shown in Fig.1.

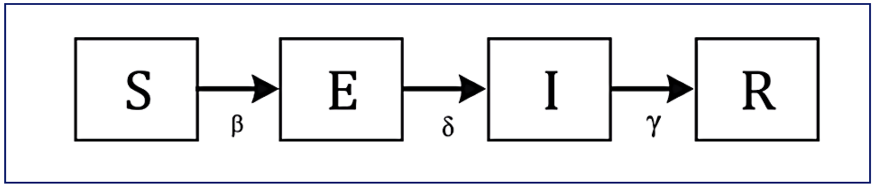
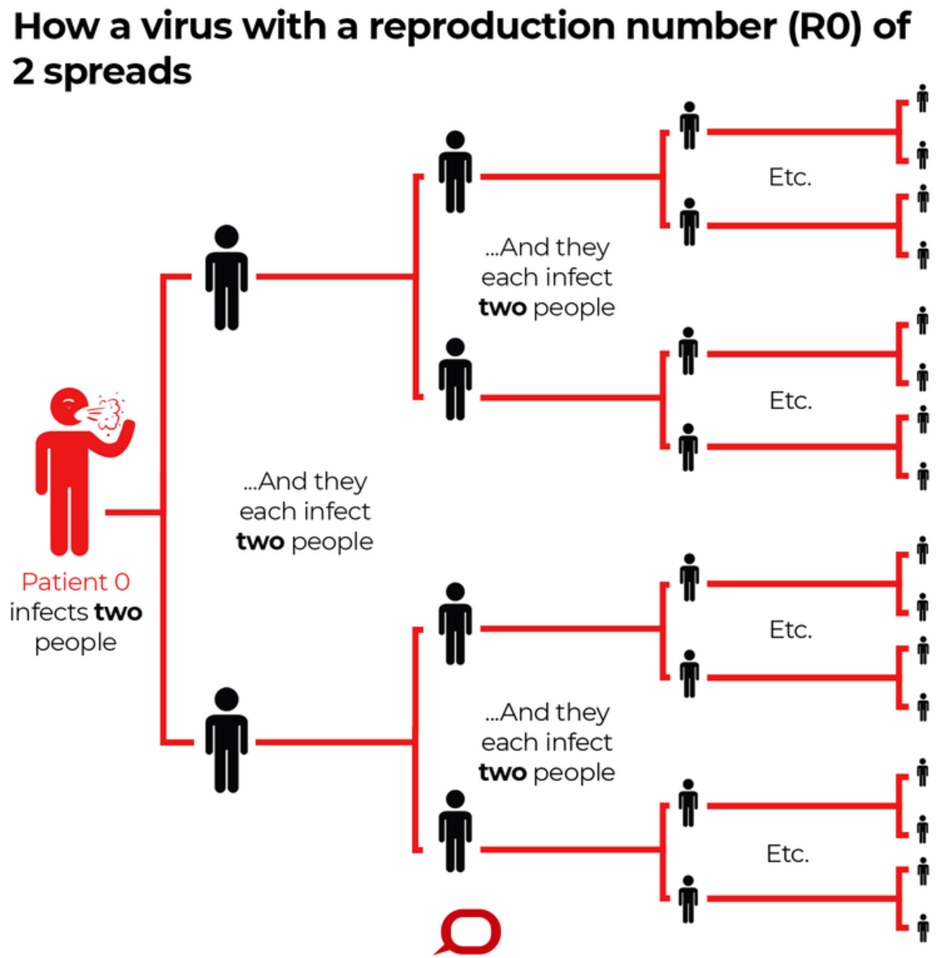


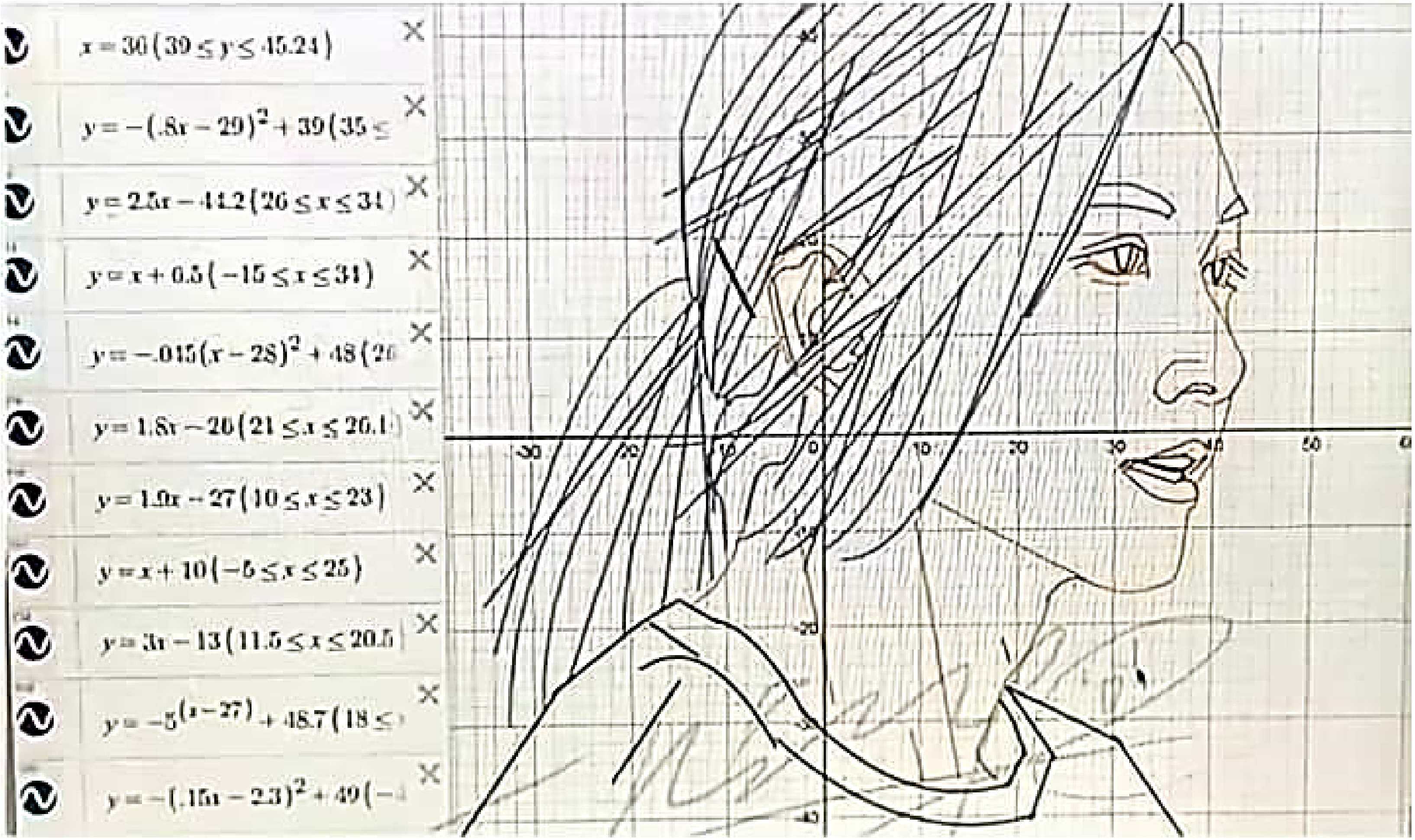
Fig 1. The Susceptible (S), Exposed (E), Infectious (I) and Removed (R) (SEIR) model



- R_0, pronounced "R naught," is a mathematical term that indicates how contagious an infectious disease is. It's also referred to as the reproduction number. As an infection is transmitted to new people, it reproduces itself.

http://covid-19.moh.gov.my/kajian-dan-penyelidikan/apa-itu-nilai-r

Artwork via equations (ctto)

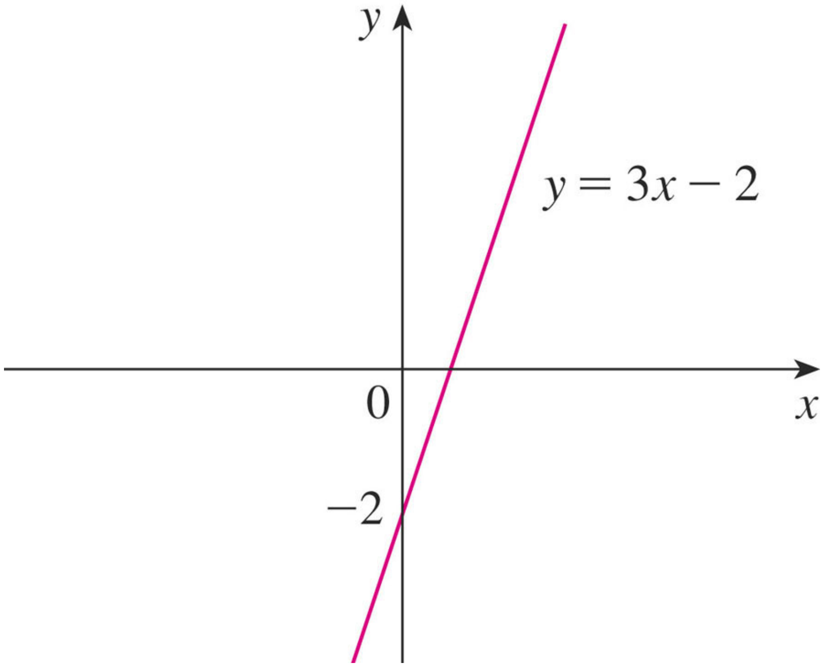


Remarks about Modeling

- All mathematical models are <u>idealizations</u>; none is ever completely accurate.
- A good model…
  - <u>simplifies</u> reality enough to permit mathematical calculations, but…
  - is <u>accurate</u> enough to provide valuable conclusions.
- There are several types of functions used in modeling, which we now discuss.

(1) Linear model

- the graph of the function is a **line**.
- $y = f(x) = mx + c$ where m is the slope of the line and c is the y-intercept.
- **Domain:** $(-\infty, \infty)$
- **Range:** $(-\infty, \infty)$



Example

T = -10h + 20

represents the rate of change of temperature in Celsius (T) with respect to height in km (h).

At a height of h = 2.5 km, the temperature is

$T = -10(2.5) + 20 = -5°C$.

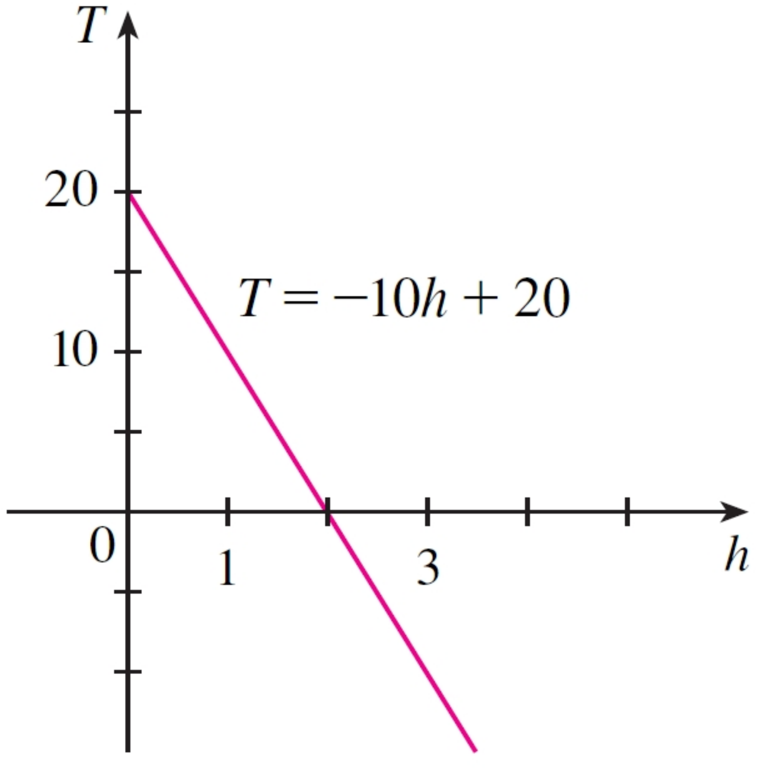


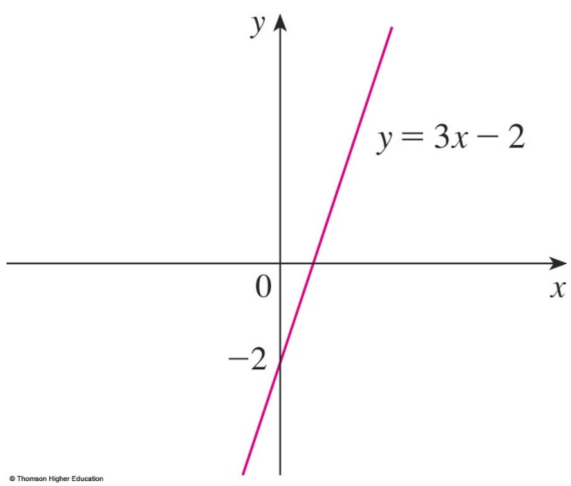
FIGURE 3

Example

For instance, the figure shows a graph of the linear function $f(x) = 3x - 2$ and a table of sample values.

– Notice that, whenever x increases by 0.1, the value of $f(x)$ increases by 0.3.

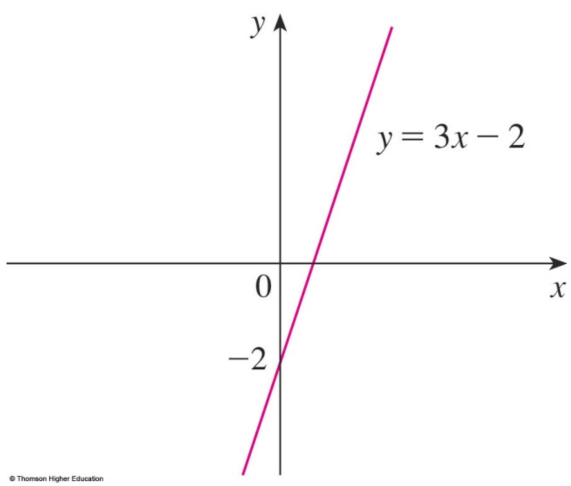
– So, $f(x)$ increases three times as fast as x.



x	$f(x) = 3x - 2$
1.0	1.0
1.1	1.3
1.2	1.6
1.3	1.9
1.4	2.2
1.5	2.5

Example (Cont')

– Thus, the slope of the graph $y = 3x - 2$, namely 3, can be interpreted as the rate of change of y with respect to x.



x	$f(x) = 3x - 2$
1.0	1.0
1.1	1.3
1.2	1.6
1.3	1.9
1.4	2.2
1.5	2.5

(2) Polynomials

- $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$ where n is a nonnegative integer and the numbers $a_0, a_1, a_2, \dots, a_n$ are constants called the coefficients of the polynomial.

- The domain of any polynomial is $\Re = (-\infty, \infty)$.

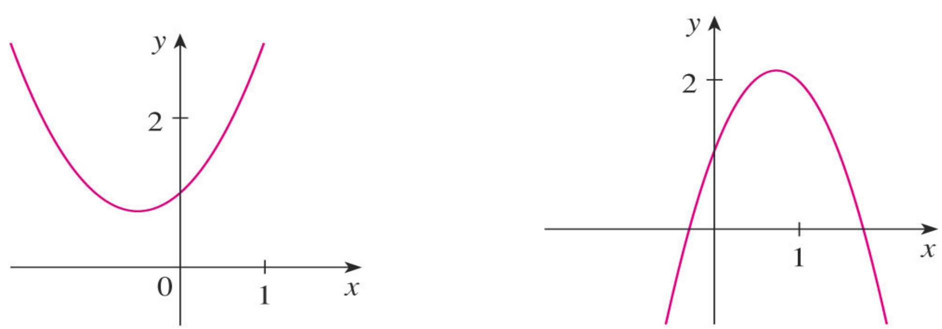
- If the leading coefficient $a_n \neq 0$, then the degree of the polynomial is n.

(2) Polynomials(Cont')

***Example* 1.10**

$P(x) = 2x^6 - x^4 + \frac{2}{5}x^3 + \sqrt{2}$ → The degree of the $P(x) = 6$.

- A polynomial of degree 1 is of the form $P(x) = mx + b$ which is linear function.
- A polynomial of degree 2 is of the form $P(x) = ax^2 + bx + c$ which is quadratic function.



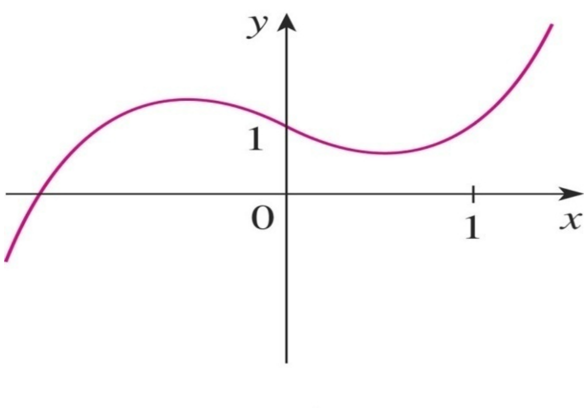
(a) $y = x^2 + x + 1$ (b) $y = -2x^2 + 3x + 1$

- For polynomial of degree 2, its graph is always a parabola obtained by shifting the parabola $y = ax^2$. The parabola opens upward if $a > 0$ and downward if $a < 0$.

(2) Polynomials(Cont')

- A polynomial of degree 3 is of the form $P(x) = ax^3 + bx^2 + cx + d$ it is called a cubic function.

- The graph shows examples of degree 4 and degree 5 polynomials. The degree gives the maximum number of "*ups and downs*" that the polynomial can have and also the maximum number of crossings of the x axis that it can have.



(a) $y = x^3 - x + 1$

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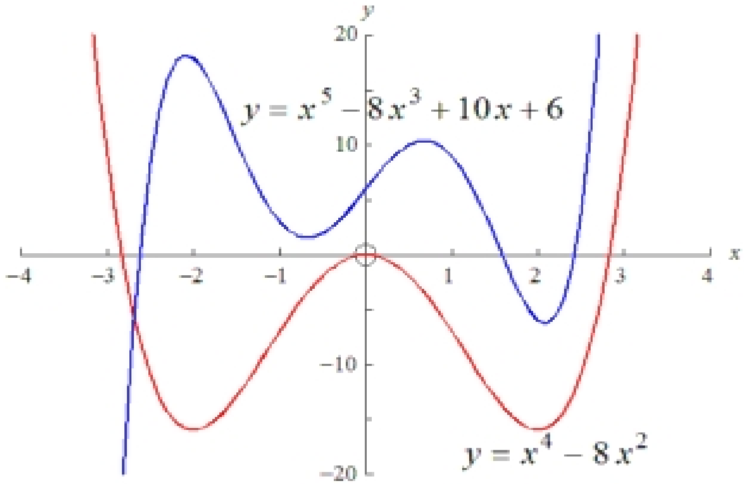
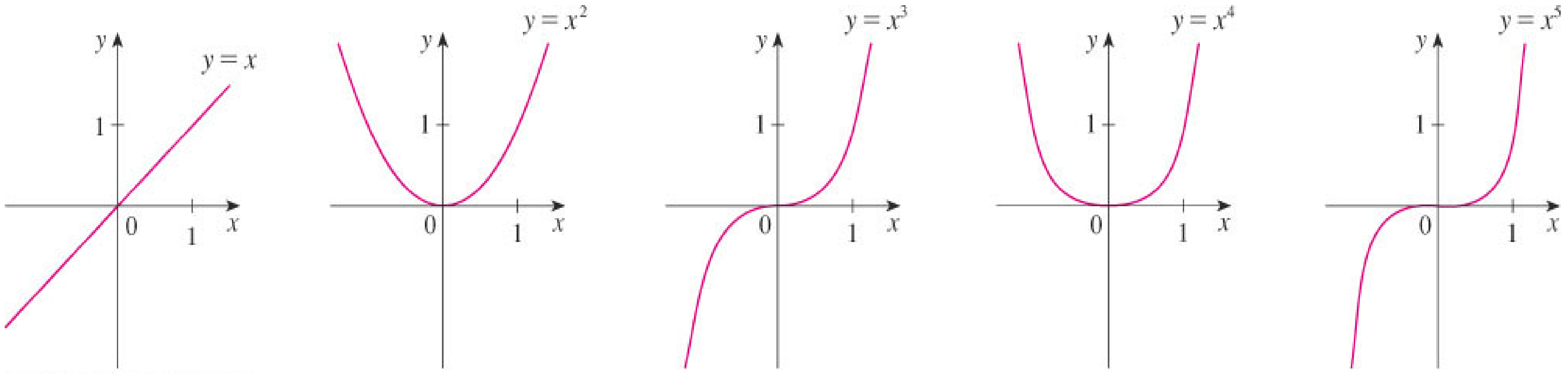


(3) Power function

- A function of the form $f(x) = x^a$, where a is a constant, is called a power function.

- **Case 1:** $a = n$, where n is a positive integer
- The graph of $f(x) = x^n$, for $n = 1, 2, 3, 4$ and 5 are shown as follows:



$n = 1$ $n = 2$ $n = 3$ $n = 4$ $n = 5$

(3) Power function (Cont')

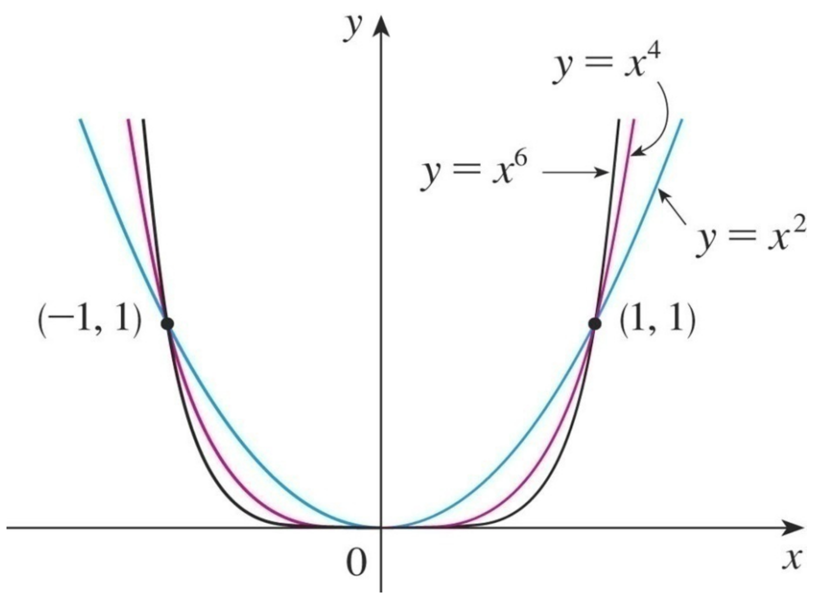
Note:

- From case 1, the general shape of the graph of $f(x) = x^n$ depends on whether n is even or odd.
- If n is even, then $f(x) = x^n$ is an even function and its graph is similar to the parabola $y = x^2$.
- If n is odd, then $f(x) = x^n$ is an odd function and its graph is similar to that of $y = x^3$.
- As n increases, the graph of $y = x^n$ becomes flatter near 0 and steeper .

(3) Power function (Cont')

***Example* 1.11**

Sketch $y = x^2$, $y = x^4$, $y = x^6$ in one graph.

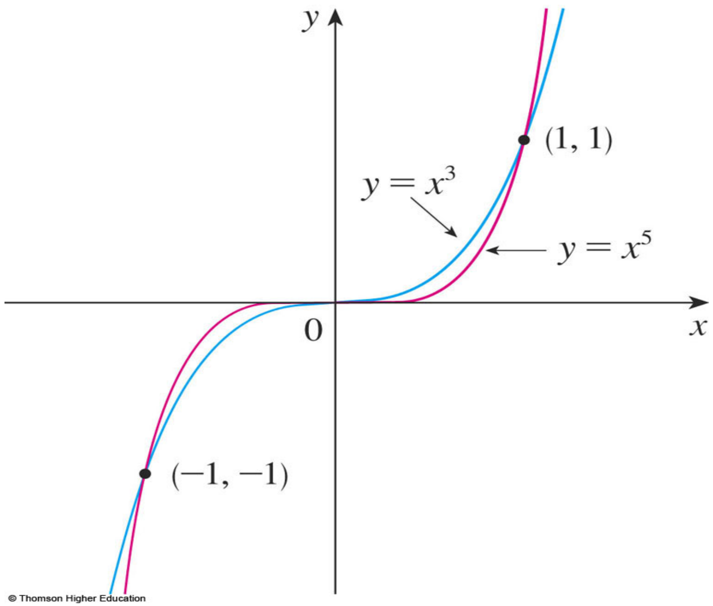


If n is even, then $f(x) = x^n$ is an even function, and its graph is similar to the parabola $y = x^2$.

(3) Power function (Cont')

***Example* 1.12**

Sketch $y = x^3$, $y = x^5$ in one graph.

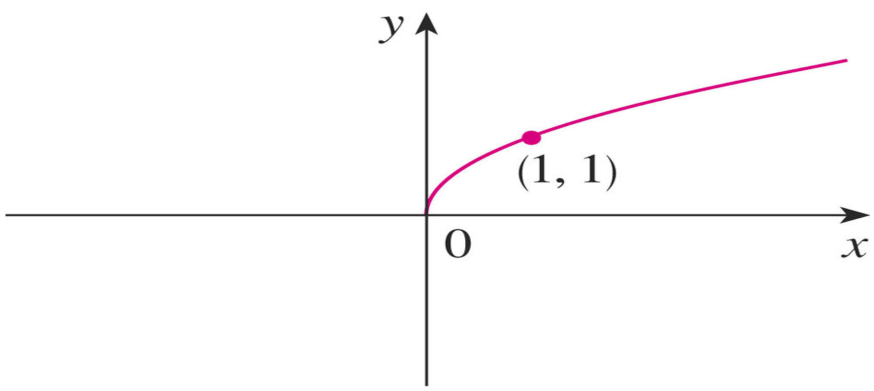


If n is odd, then $f(x) = x^n$ is an odd function, and its graph is similar to that of $y = x^3$.

(3) Power function (Cont')

Case 2: $a = 1/n$, where n is a positive integer

- The function $f(x) = x^{\frac{1}{n}} = \sqrt[n]{x}$ is a root function.
- For $n = 2$, $f(x) = \sqrt{x}$ is the square root function.

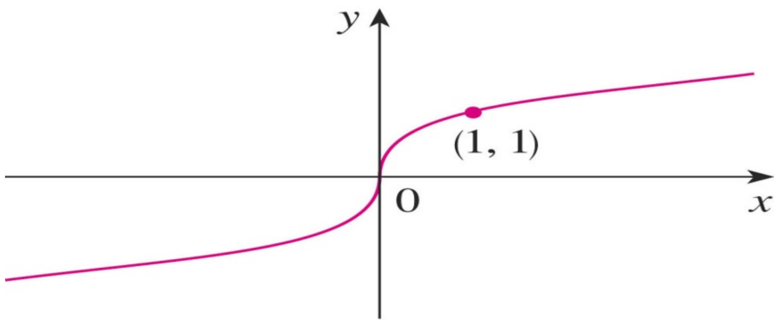


(a) $f(x) = \sqrt{x}$

(3) Power function (Cont')

Case 2: a =1/n, where n is a positive integer (Cont')

- For n = 3, $f(x) = \sqrt[3]{x}$.



(b) $f(x) = \sqrt[3]{x}$

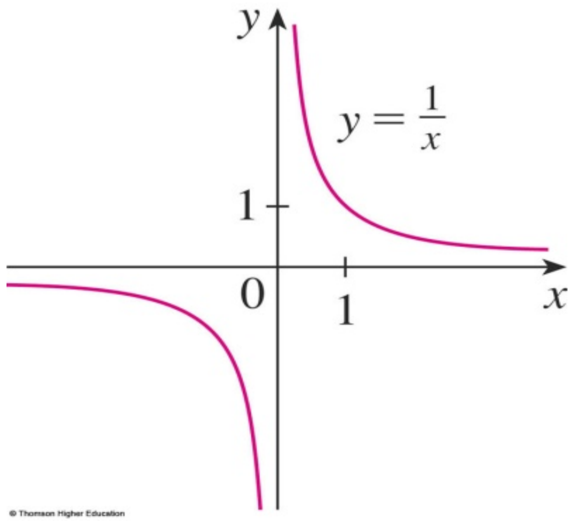
Note:

- The graph of $y = \sqrt[n]{x}$ for n is odd (n >3) is similar to that of $y = \sqrt[3]{x}$
- The graph $y = \sqrt[n]{x}$ for n is even (n >2) is similar to that of $y = \sqrt{x}$.

(3) Power function (Cont')

Case 3: a = -1

- The graph of the reciprocal function $f(x) = \dfrac{1}{x}$ is shown as follows:



- Note:
- Since $x \neq 0$, thus y-axis ($x = 0$) is a vertical asymptote.
- Since $y = 1/x$, $x = 1/y$, thus $y \neq 0$, thus $y = 0$ (x-axis) is a horizontal asymptote.

(4) Rational functions

(4) Rational functions

- A rational function *f* is a ratio of two polynomials:

$$f(x) = \frac{P(x)}{Q(x)}$$

- where *P* and *Q* are polynomials.
- The domain consist of all values of *x* such that $Q(x) \neq 0$.

(4) Rational functions (Cont')

(4) Rational functions (Cont')

***Example* 1.13**

Find the domain of the following polynomials:

(a) $f(x) = 1/x$

A simple example of a rational function is the function $f(x) = 1/x$, whose domain is $\{x \mid x \neq 0\}$.

(b) $f(x) = \dfrac{2x^4 - x^2 + 1}{x^2 - 4}$

The function is a rational function with domain $\{x \mid x \neq \pm 2\}$.

(5) Algebraic functions

(5) Algebraic functions

- A function *f* is called an algebraic function if it can be constructed using algebraic operations (such as addition, subtraction, multiplication, division, and taking roots) starting with polynomials.

Note:

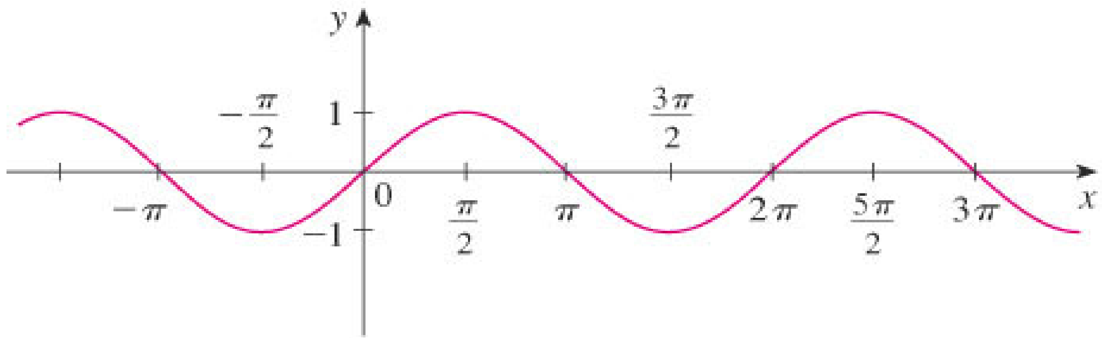
- Rational function is an algebraic function.
- But algebraic function is not necessary a rational function.

***Example* 1.14**

$$f(x)=\sqrt{x^2+1} \qquad g(x)=\frac{x^4-16x^2}{x+\sqrt{x}}+(x-2)\sqrt[3]{x+1}$$

(6) Trigonometric functions

(i) $y = \sin x$, $x \in [0, 2\pi]$, range $\in$ [-1,1]

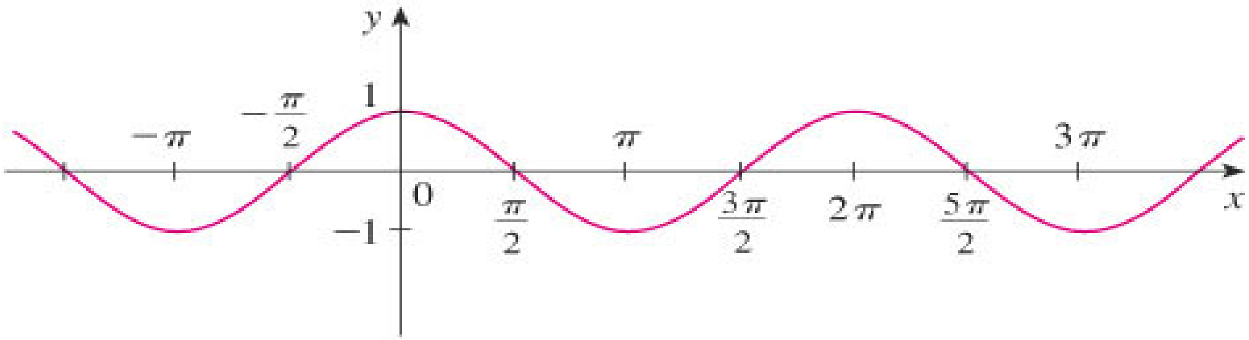


(a) $f(x) = \sin x$

- Note:
- $y = \sin x = \sin(x + 2\pi)$ as the graph is periodic over a period of 2π.
- $\sin(-x) = -\sin x$

(6) Trigonometric functions (Cont')

(i) $y = \cos x$, $x \in [0, 2\pi]$, range $\in$ [-1,1]

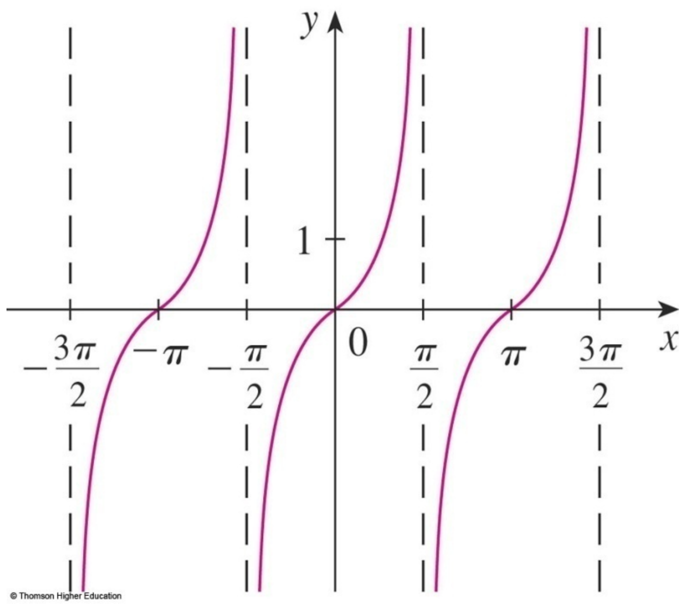


(b) $g(x) = \cos x$

- Note:
- $y = \cos x = \cos(x + 2\pi)$ as the graph is periodic over a period of 2π.
- $\cos(-x) = \cos x$

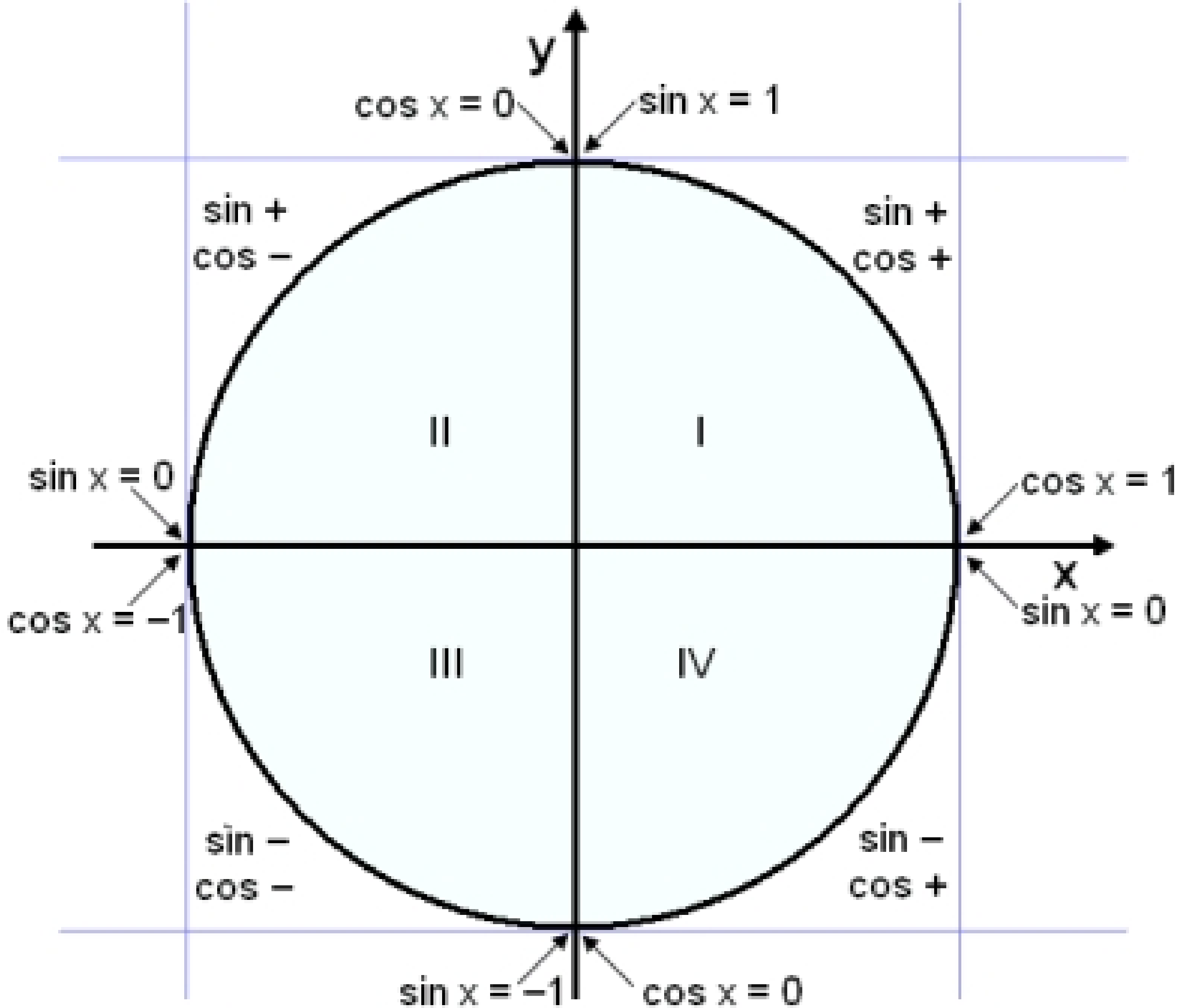
(6) Trigonometric functions (Cont')

(iii) $y = \tan x = \dfrac{\sin x}{\cos x}, \cos x \neq 0$ (means $x \neq \dfrac{n\pi}{2}$, n is nonzero integer)



Note:

$$\tan(x+\pi) = \tan x$$


y
cos x = 0
sin x = 1
sin +
cos −
sin +
cos +
II
I
sin x = 0
cos x = 1
x
cos x = −1
sin x = 0
III
IV
sin −
cos −
sin −
cos +
sin x = −1
cos x = 0

Trigonometry Ratio of specific angles

	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$cosec\,\theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot\theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

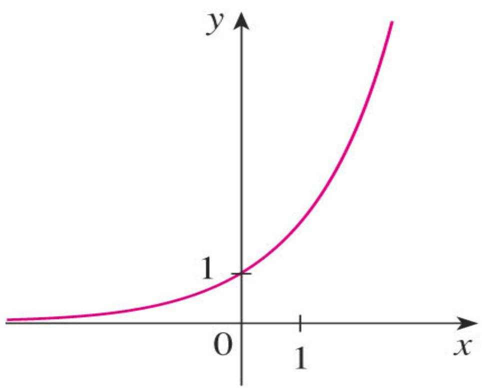
(7) Exponential functions

A function f of the form $f(x) = a^x$, where $a > 0$.

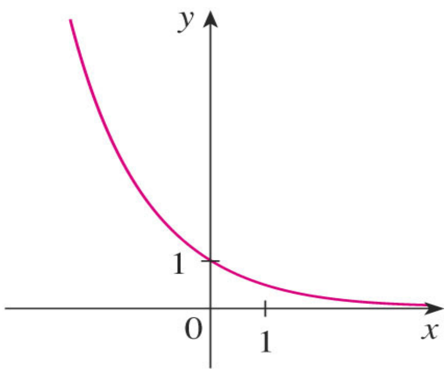
- ***Example* 1.15**

(a) $f(x) = 2^x$

(b) $f(x) = \left(\dfrac{1}{2}\right)^x$, $0 < a < 1$



(a) $y = 2^x$



(b) $y = (0.5)^x$

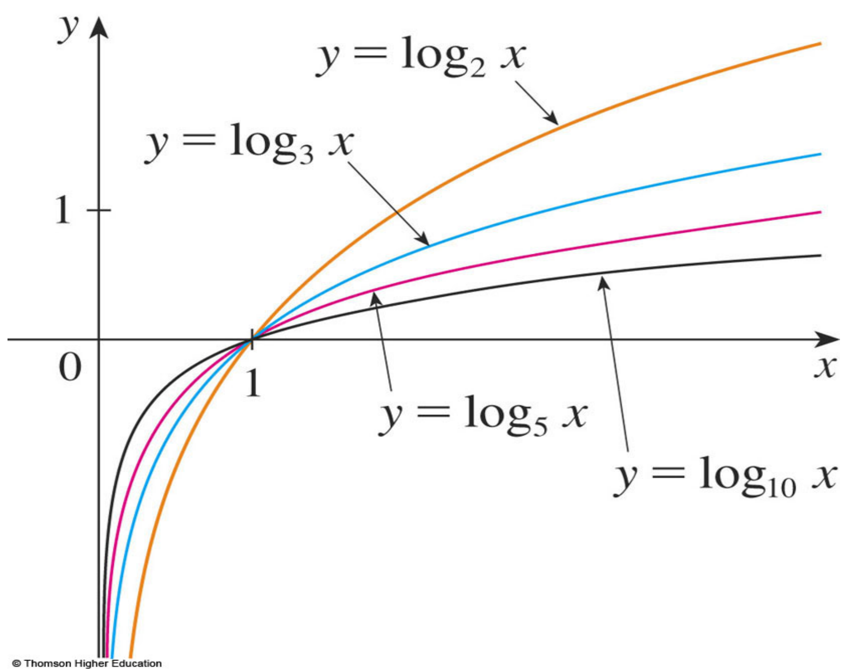
For both cases, the domain is $(-\infty, \infty)$ and the range is $(0, \infty)$.

(8) Logarithmic functions

- $\boxed{f(x)=\log_a x}$,where $a > 0$

***Example* 1.16**

Sketch $y=\log_2 x,\ y=\log_3 x,\ y=\log_5 x,\ y=\log_{10} x$ in one graph. State the domain and range.



In each case, the domain is $(0,\infty)$ the range is $(-\infty,\infty)$, and the function increases slowly when $x > 1$.

(9) Transcendental functions

- These are functions that are not algebraic.
- Example of the transcendental functions is trigonometric, inverse trigonometric, exponential, and logarithmic functions.

***Example* 1.17**

$y = \sin^{-1} x$, $y = \text{cosec } x$, $y = e^x$, $y = \ln x$

(9) Transcendental functions (Cont')

Classify the following functions as one of the types of functions that we have discussed.

a. $f(x) = 5^x$

b. $g(x) = x^5$

c. $h(x) = \dfrac{1+x}{1-\sqrt{x}}$

d. $u(t) = 1 - t + 5t^4$

(9) Transcendental functions (Cont')

$f(x) = 5^x$ is an exponential function.

- The x is the exponent.

$g(x) = x^5$ is a power function.

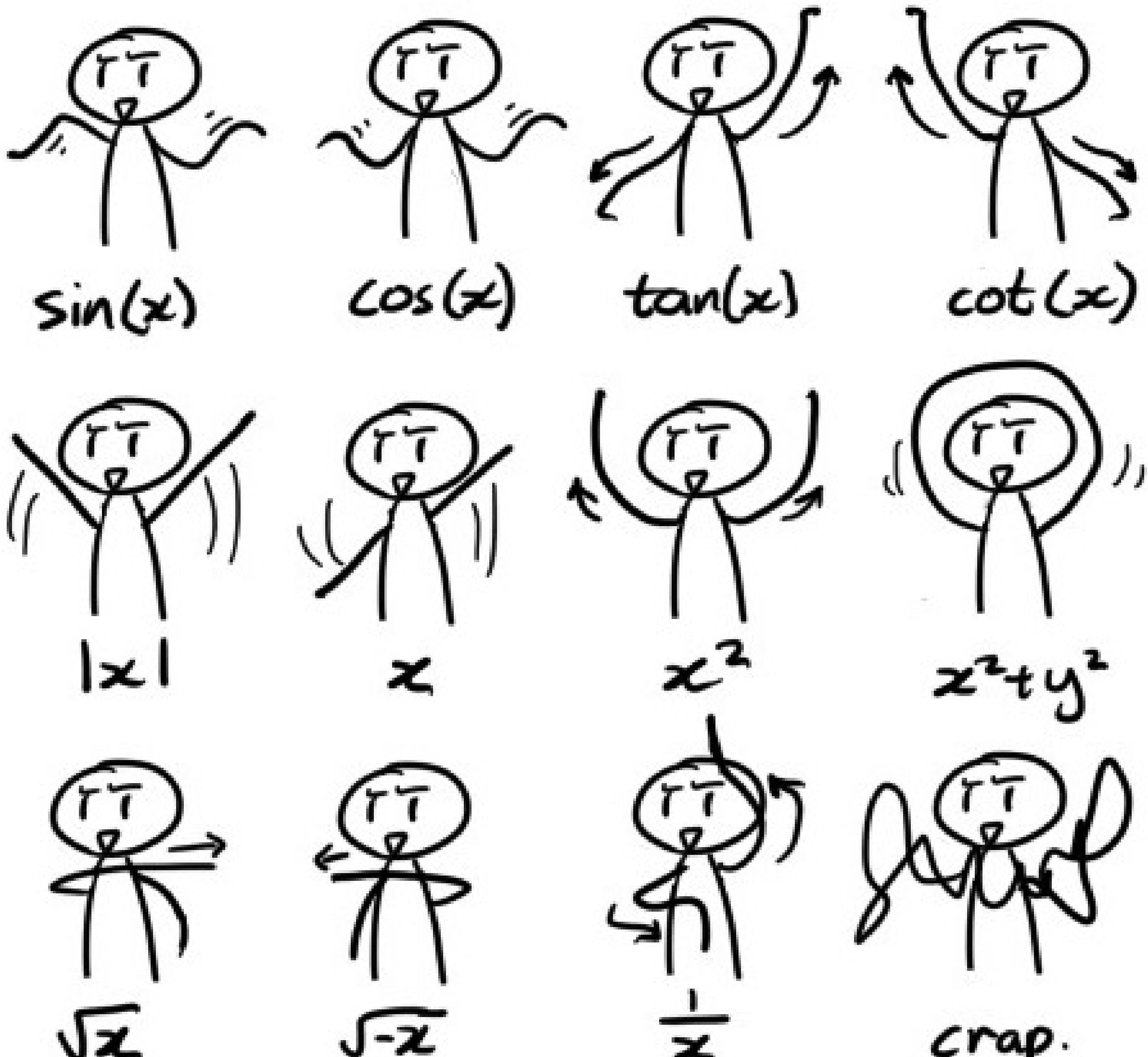
- The x is the base.

We could also consider it to be a polynomial of degree 5.

(9) Transcendental functions (Cont')

$h(x) = \dfrac{1+x}{1-\sqrt{x}}$ is an algebraic function.

$u(t) = 1 - t + 5t^4$ is a polynomial of degree 4.



sin(x)
cos(x)
tan(x)
cot(x)
|x|
x
x²
x²+y²
√x
√-x
1/x
crap.

Section 1.3

Transformations, combinations, composition and graphs of functions

(1) Transformation of functions

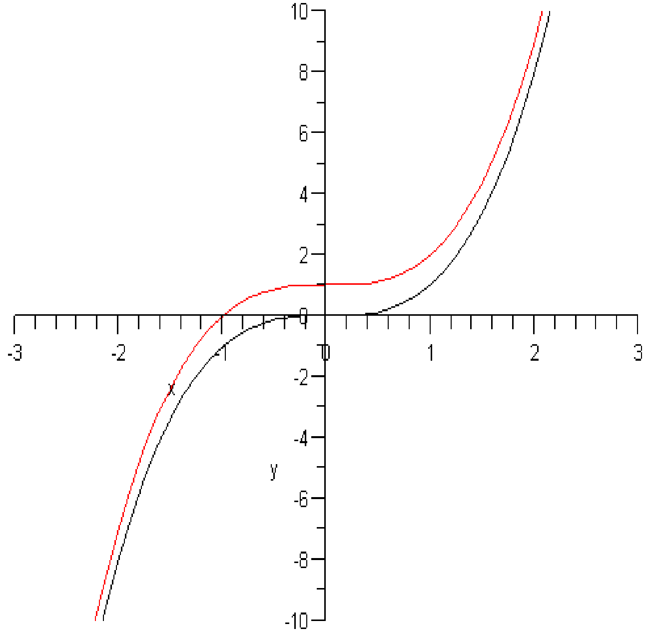
(a) Translation

- Moving the whole graph without changing its shape.

(i) Suppose $c > 0$, to obtain the graph of $y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units upward (**up to y-axis**).

Example **1.18**

Sketch $y = x^3+1$

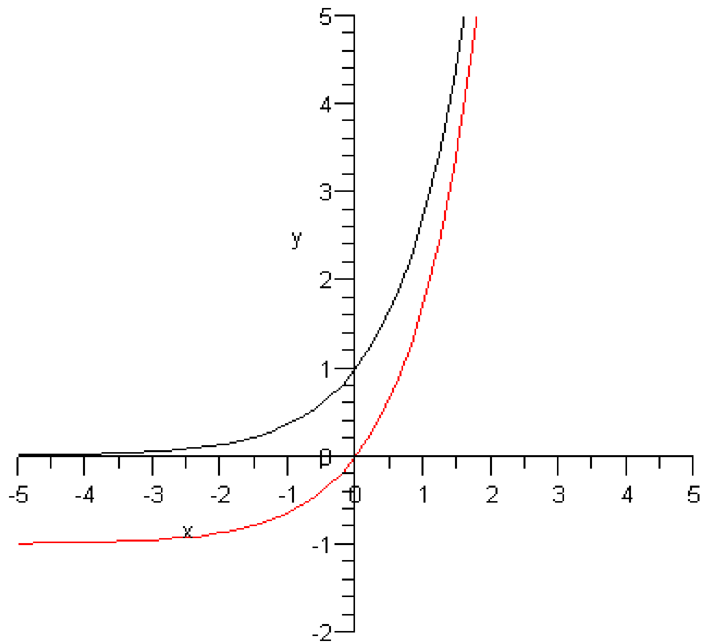


(a) Translation (Cont')

(ii) Suppose $c > 0$, to obtain the graph of $y = f(x) \boxed{- c}$, shift the graph of $y = f(x)$ a distance c units downward (down from y-axis).

***Example* 1.19**

Sketch $y = e^x - 1$.

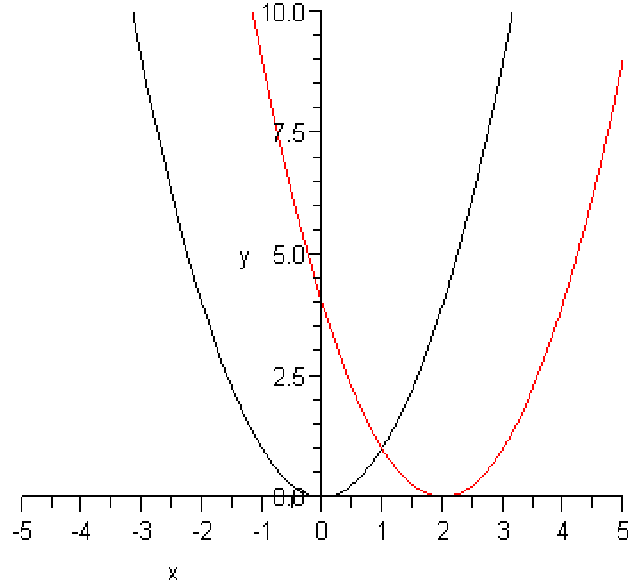


(a) Translation (Cont')

(iii) Suppose $c > 0$, to obtain the graph of $y = f(\boxed{x - c})$, shift the graph of $y = f(x)$ a distance c units to the right of x-axis.

***Example* 1.20**

Sketch $y = (x - 2)^2$

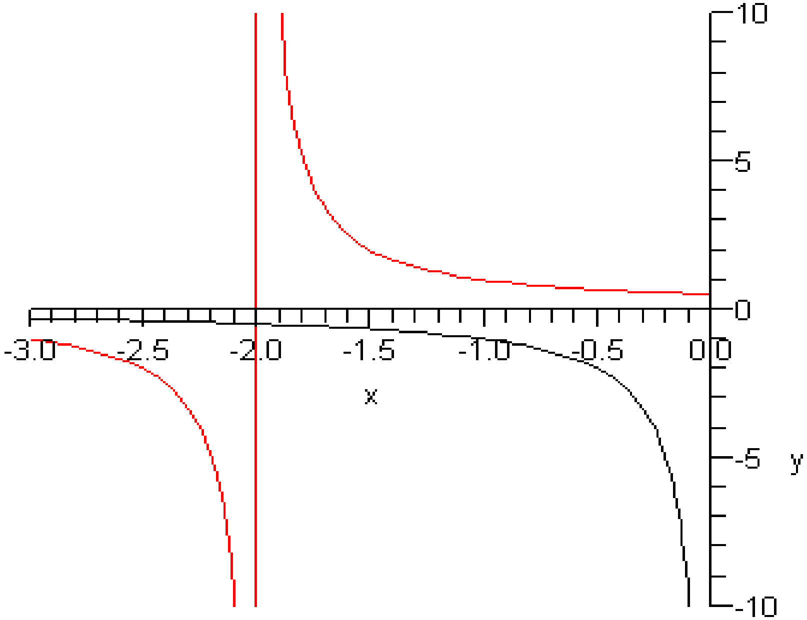


(a) Translation (Cont')

(iv) Suppose $c > 0$, to obtain the graph of $y = f(x \boxed{+ c})$, shift the graph of $y = f(x)$ a distance c units to the left of x-axis.

<u>*Example* 1.21</u>

Sketch $y = \dfrac{1}{x+2}$.



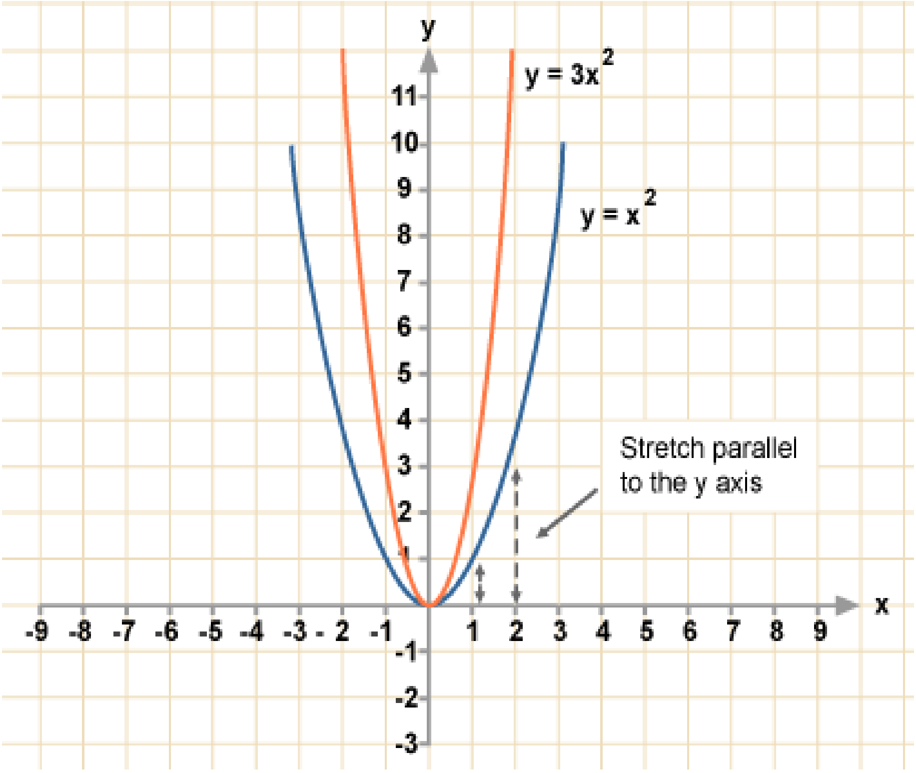
(b) Stretching

- Consists of stretching and compressing horizontally or vertically.

(i) Suppose $c > 1$, the graph of $y = c\,f(x)$ can be obtained by stretching the graph $y = f(x)$ vertically by a factor of c.

***Example* 1.22**

Sketch $y = 3x^2$.

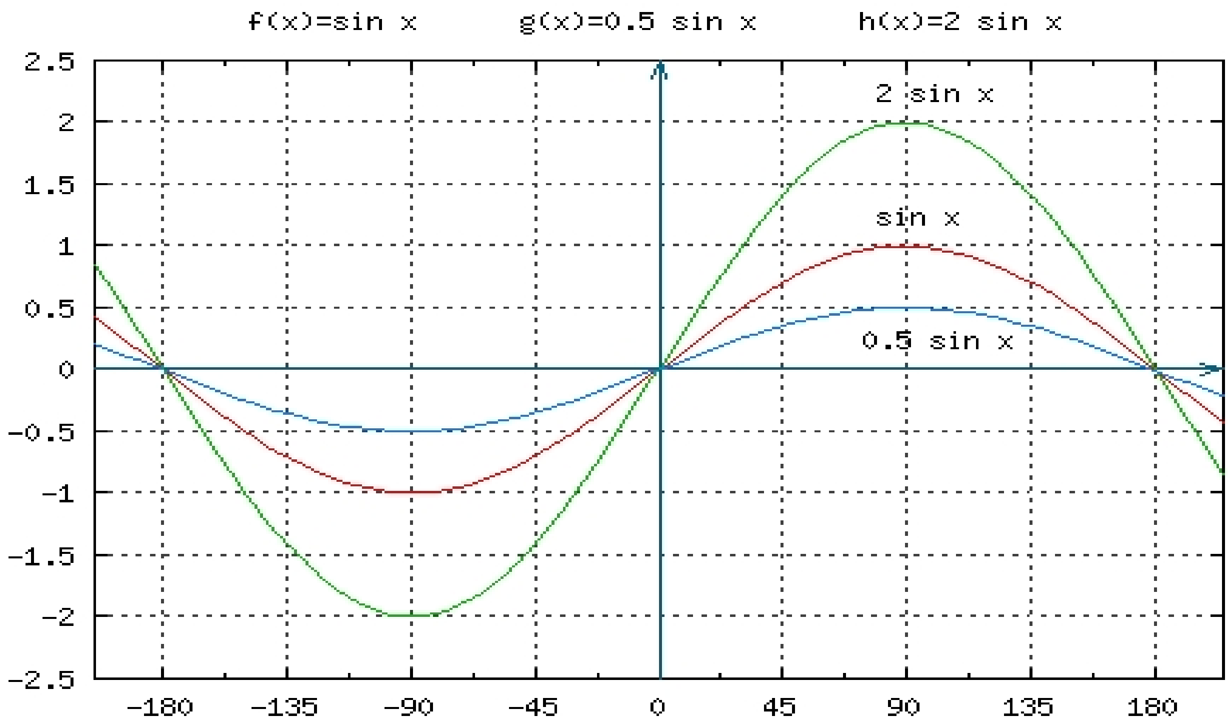


(b) Compressing (Cont')

(ii)Suppose $c > 1$, the graph of $\boxed{y = \frac{1}{c} f(x)}$ can be obtained by compressing the graph $y = f(x)$ vertically by a factor of c.

***Example* 1.23**

Sketch $y = \frac{1}{2} \sin x$(vertical compress by a factor of 0.5)

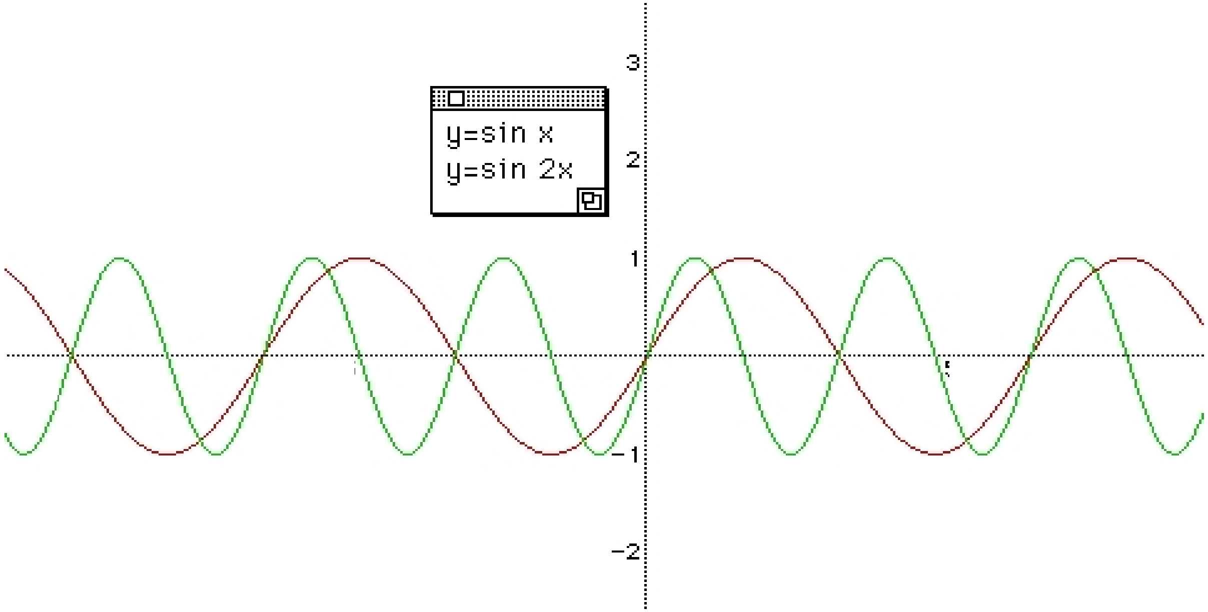


(b) Compressing (Cont')

- Suppose $c > 1$, the graph of $y = f\,(cx)$ can be obtained by compressing the graph $y = f\,(x)$ horizontally by a factor of c.

***Example* 1.24**

Sketch $y = \sin\,(2x)$.

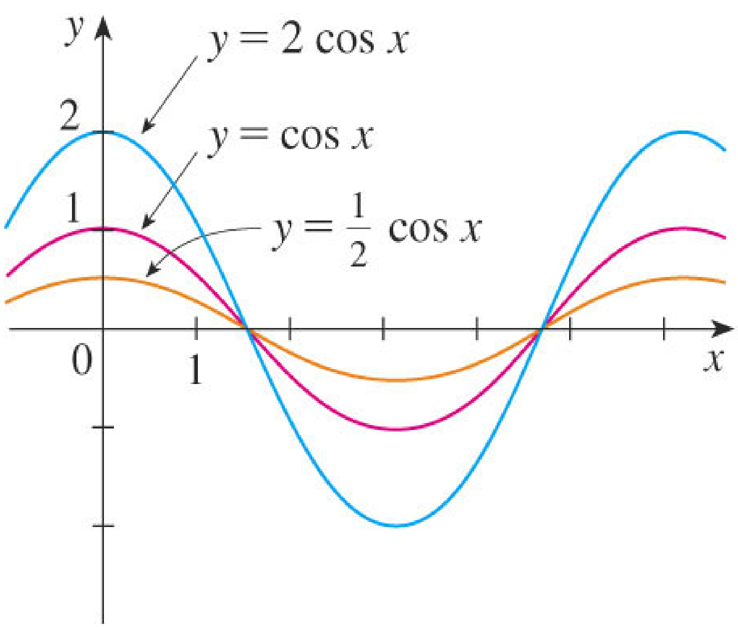


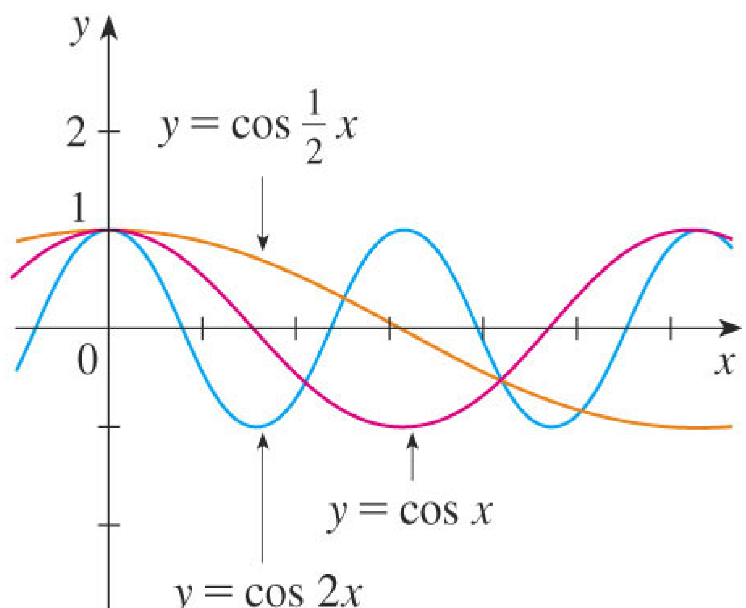
(b) Stretching (Cont')

- Suppose $c > 1$, the graph of $y = f\left(\frac{x}{c}\right)$ can be obtained by stretching the graph $y = f(x)$ horizontally by a factor of c.

<u>*Example* 1.25</u>

Sketch $y = \cos\left(\frac{x}{2}\right)$.



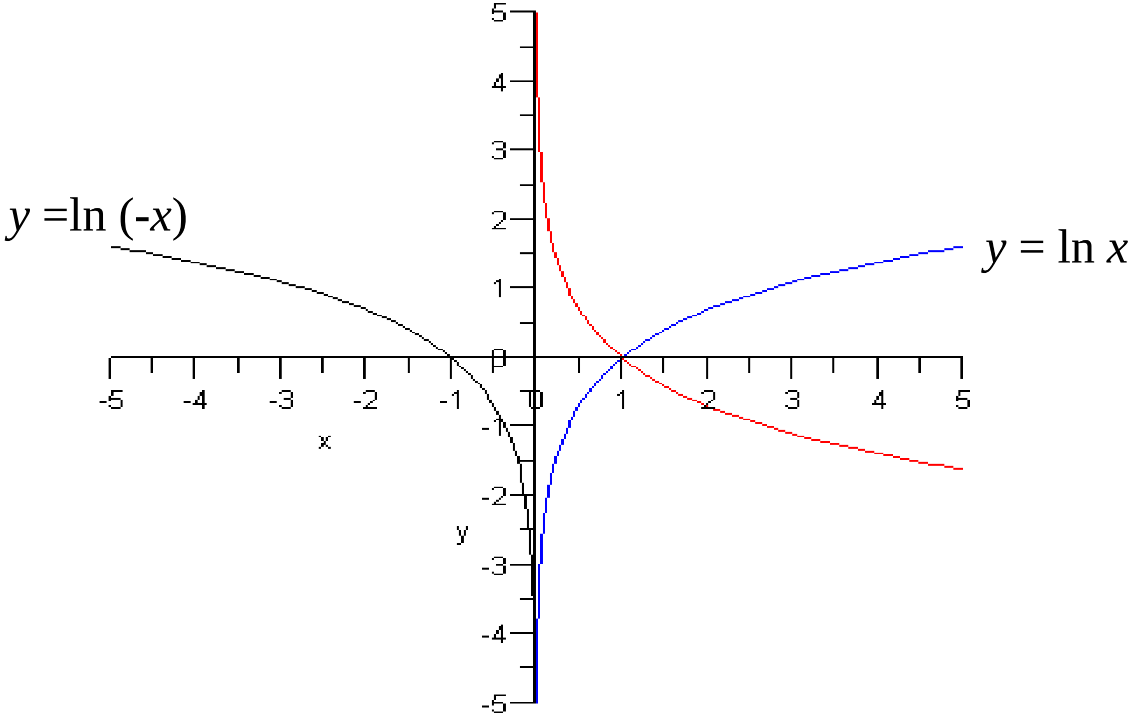


(c) Reflecting (Cont')

- Suppose $c > 1$, the graph of $y = f\,(-x)$ can be obtained by reflecting the graph $y = f\,(x)$ about the y-axis.

***Example* 1.26**

Sketch $y = \ln(-x)$.

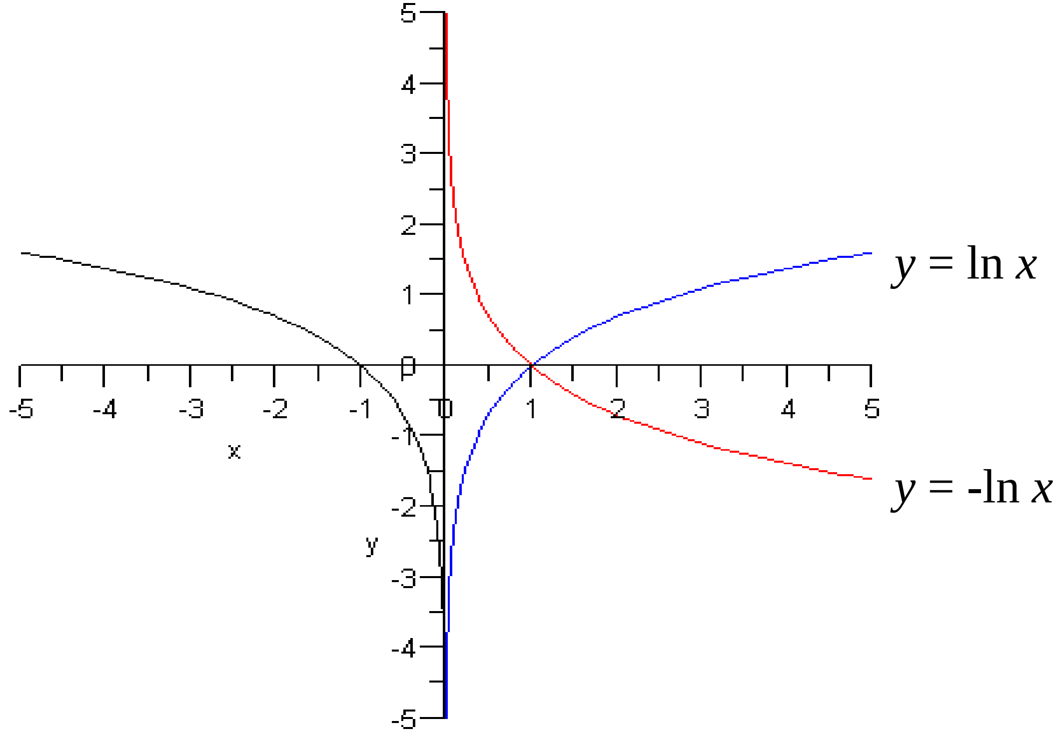


(c) Reflecting (Cont')

- Suppose c > 1, the graph of y = $-f(x)$ can be obtained by reflecting the graph $y = f(x)$ about the x-axis.

***Example* 1.27**

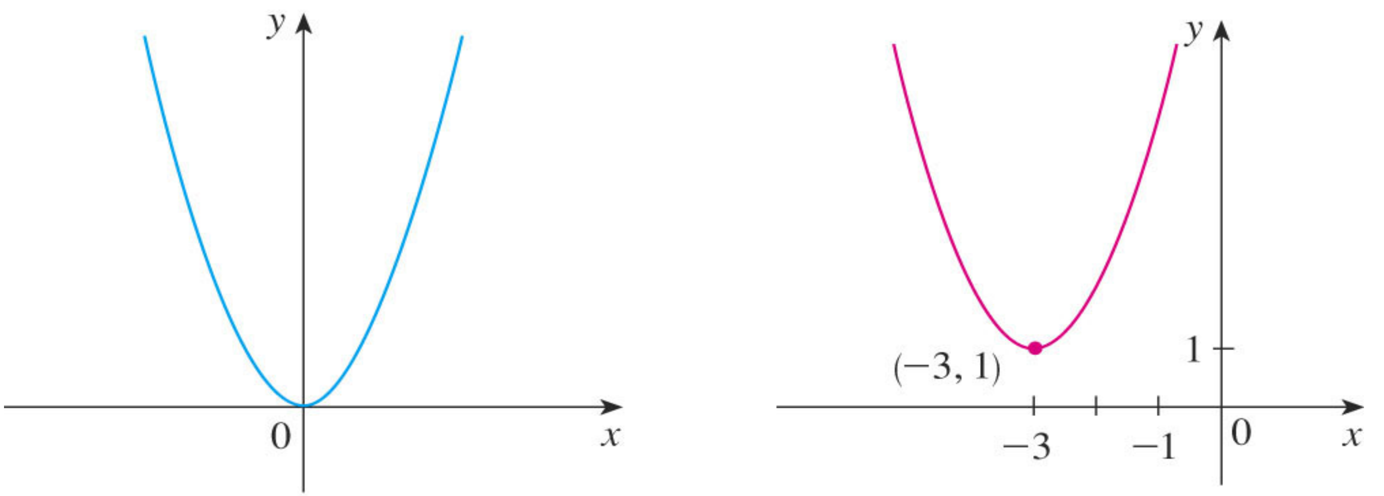
Sketch $y = -\ln x$.



***Example* 1.28**

Sketch the graph of the function

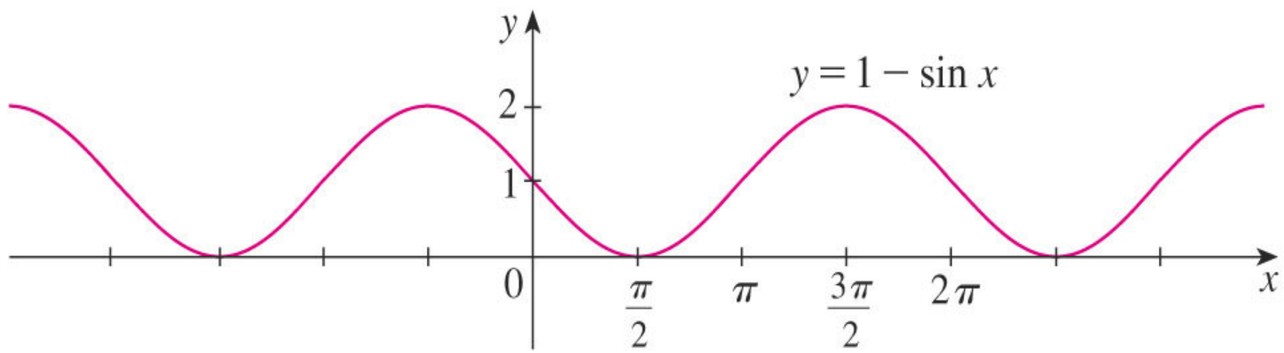
$$f(x) = x^2 + 6x + 10$$



(a) $y = x^2$ (b) $y = (x+3)^2 + 1$

***Example* 1.29**

Sketch the graph of the function $f(x) = 1 - \sin x$.



Question (TRY)

Given the graph of $y = \sqrt{x}$, use transformations to graph:

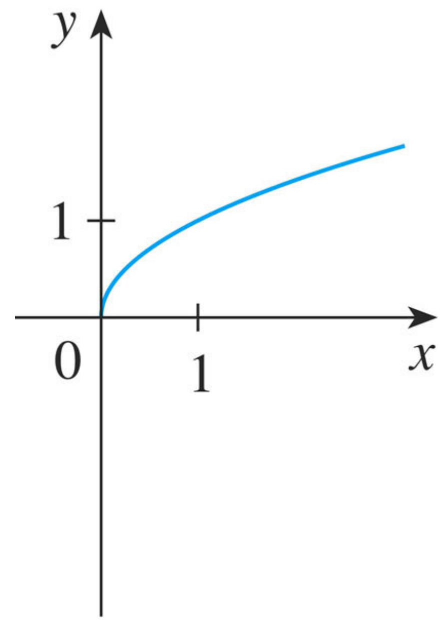
a. $y = \sqrt{x} - 2$

b. $y = \sqrt{x - 2}$

c. $y = -\sqrt{x}$

d. $y = 2\sqrt{x}$

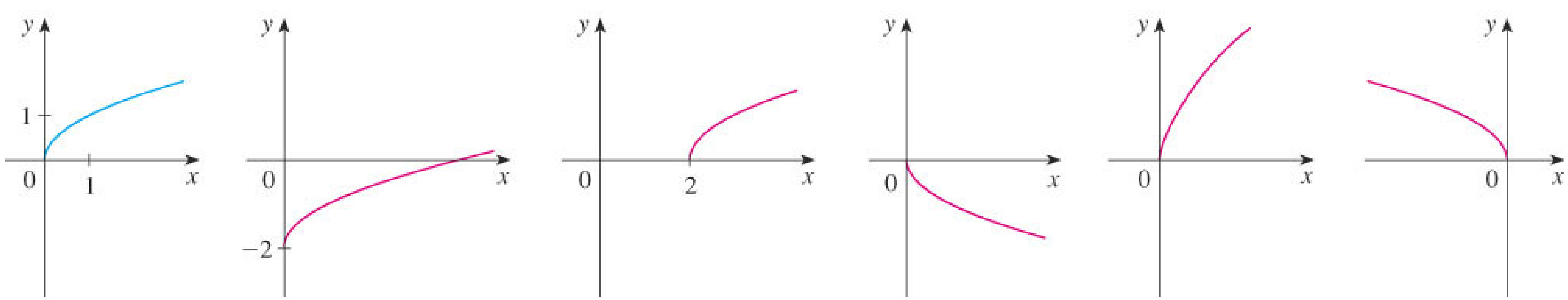
e. $y = \sqrt{-x}$



(a) $y = \sqrt{x}$

Question (TRY)) (Cont')

- In the other parts of the figure, we sketch:
  - $y = \sqrt{x} - 2$ by shifting 2 units downward.
  - $y = \sqrt{x - 2}$ by shifting 2 units to the right.
  - $y = -\sqrt{x}$ by reflecting about the *x*-axis.
  - $y = 2\sqrt{x}$ by stretching vertically by a factor of 2.
  - $y = \sqrt{-x}$ by reflecting about the *y*-axis.



(a) $y = \sqrt{x}$ (b) $y = \sqrt{x} - 2$ (c) $y = \sqrt{x - 2}$ (d) $y = -\sqrt{x}$ (e) $y = 2\sqrt{x}$ (f) $y = \sqrt{-x}$

(2) Combinations of functions

- Let *f* and *g* be functions with domains *A* and *B*. Then the functions *f* + *g*, *f* - *g*, *fg*, and are defined as follows:

➢ $(f + g)(x) = f(x) + g(x)$, domain = $A \cap B$

➢ $(f - g)(x) = f(x) - g(x)$, domain = $A \cap B$

➢ $(fg)(x) = f(x)g(x)$, domain = $A \cap B$

➢ $\left(\dfrac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)}$, domain = $\{x \in A \cap B \mid g(x) \neq 0\}$

(2) Combinations of functions (Cont')

***Example* 1.30**

Given $f(x)=\sqrt{x}$ and $g(x)=\sqrt{2-x}$, find the functions $f+g$, $f-g$, fg, and f/g and their respective domains.

For example, the domain of $f(x)=\sqrt{x}$ is $A=[0,\infty)$ and the domain of $g(x)=\sqrt{2-x}$ is $B=(-\infty,2]$. The interception of both domains of A and B is $[0,\infty)\cap(-\infty,2]=[0,2]$.

So, the domain of $(f+g)(x)=\sqrt{x}+\sqrt{2-x}$ is $A\cap B=[0,2]$.

the domain of $(f-g)(x)=\sqrt{x}-\sqrt{2-x}$ is $A\cap B=[0,2]$.

the domain of $(fg)(x)=\sqrt{x}\sqrt{2-x}=\sqrt{x(2-x)}$ is $A\cap B=[0,2]$.

the domain of $\left(\frac{f}{g}\right)(x)=\frac{\sqrt{x}}{\sqrt{2-x}}=\sqrt{\frac{x}{2-x}}$ is $\{x\in A\cap B \mid g(x)\neq 0\}$ or [0,2).

(3) Composition of functions

- Given two functions f and g, the composite function f g (also called the composition of f and g) is defined by $(f \circ g)(x) = f(g(x))$.

- Suppose that $y = f(u) =$ and $u = g(x) = x^2 + 1$. Then $y = f(u) = f(g(x)) = f(x^2 + 1) = \sqrt{x^2 + 1}$.

Example

- If $f(x) = x^2$ and $g(x) = x - 3$, find the composite functions $f \circ g$ and $g \circ f$.
- Solution By definition of composition,

$$(f \circ g)(x) = f(g(x)) = f(x - 3) = (x - 3)^2$$

and

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 - 3\ .$$

- Note that in general $f \circ g \neq g \circ f$!

(3) Composition of functions (Cont')

Example **1.31**

Given $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$. Find $f \circ g$, $g \circ f$, $f \circ f$, and $g \circ g$ and their respective domains.

$$\boxed{(f \circ g)(x) = f(g(x)) = f(\sqrt{2-x}) = \sqrt{\sqrt{2-x}} = \sqrt[4]{2-x}}$$

The domain, $\{x \mid 2-x \geq 0\} = \{x \mid x \leq 2\} = (-\infty, 2]$

$$\boxed{(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{2-\sqrt{x}}}, \; x \geq 0$$

- For $\sqrt{2-\sqrt{x}}$ to be defined, we must have, $2-\sqrt{x} \geq 0$
- So, $\sqrt{x} \leq 2$ or $x \leq 4$
- As a result, domain for $g \circ f$ is $0 \leq x \leq 4$ or is the closed interval [0, 4].

(3) Composition of functions (Cont')

***Example* 1.31(Cont')**

$$(f\circ f)(x) = f(f(x)) = f(\sqrt{x}) = \sqrt{\sqrt{x}} = \sqrt[4]{x}$$

The domain for $f\circ f$ is $[0,\infty)$.

$$(g\circ g)(x) = g(g(x)) = g(\sqrt{2-x}) = \sqrt{2-\sqrt{2-x}}$$

Domain, $2-x \geq 0$ and $2-\sqrt{2-x} \geq 0$

$x \leq 2$ and $\sqrt{2-x} \leq 2$

$2-x \leq 4$

$x \geq -2$

So the domain $-2 \leq x \leq 2$ is or the closed interval of [-2,2].

Section 1.4

Exponential functions

Exponential functions

- In general, an *exponential function* is a function of the form $f(x) = a^x$, where $a > 0$ (a is a positive constant).

- If $x = n$, n is positive integer, then $a^n = \underbrace{a a \cdots a}_{n \text{ factors}}$

- $a^0 = 1$

- $a^{\frac{p}{q}} = (\sqrt[q]{a})^p$

- $a^{-n} = \dfrac{1}{a^n}$

Exponential functions (Cont')

- If x is…
  - a *rational number*, $x = p/q$, where p and q are integers and $q > 0$, then

$$a^x = a^{p/q} = \sqrt[q]{a^p} = \left(\sqrt[q]{a}\right)^p$$

  - an *irrational number*, say $x = \sqrt{3} = 1.7320\ldots$

    then we define $a^{\sqrt{3}}$ to be the number approximated by

$$2^{1.7},\ 2^{1.73},\ 2^{1.732},\ 2^{1.7320},\ 2^{1.73206},\ \ldots$$

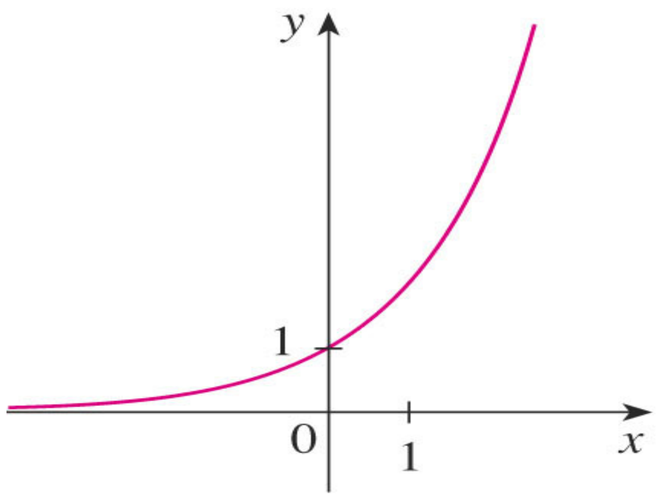
Three cases of exponential function

1) $y = a^x$, $a > 1$, $x \in \Re$

<u>*Example* **1.32**</u>

$y = 2^x$, $y = e^x$

It is an increasing function.



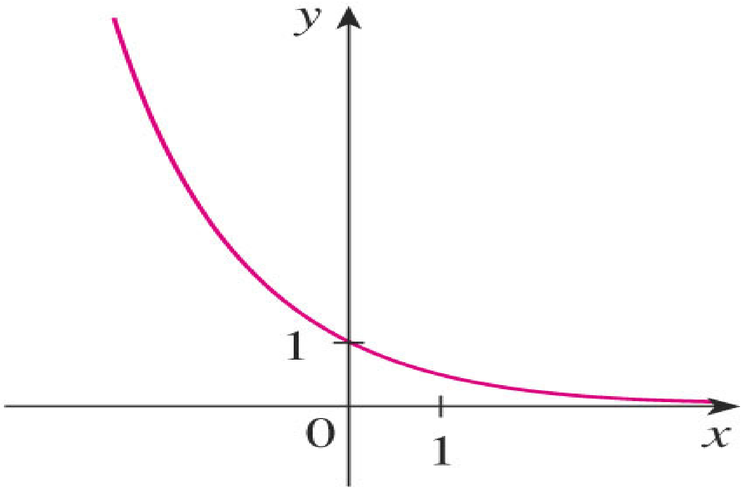
(a) $y = 2^x$

Three cases of exponential function (Cont')

2) $y = a^x$, $0 < a < 1$, $x \in \Re$

***Example* 1.33**

$y = \left(\dfrac{1}{2}\right)^x$ It is a decreasing function.

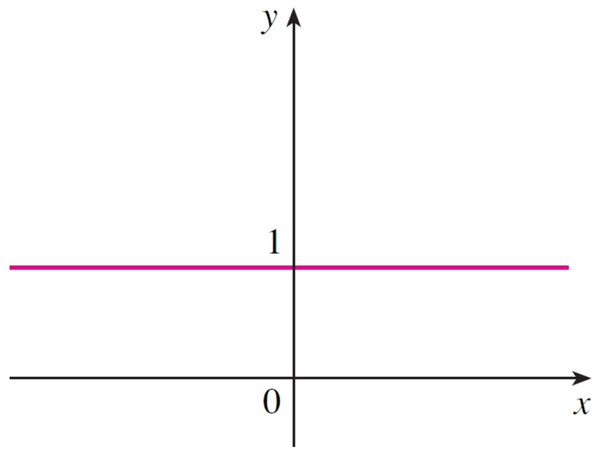


(b) $y = (0.5)^x$

Three cases of exponential function (Cont')

3) $y = a^x$, $a = 1, x \in \Re$ ⟶ $y = 1$

It is a constant function.



(b) $y = 1^x$

Note:

- Domain for $y = a^x$ is $(-\infty, \infty)$.
- Range for $y = a^x$ is $(0, \infty)$ for $a \neq 1$.

Graphs of a^x

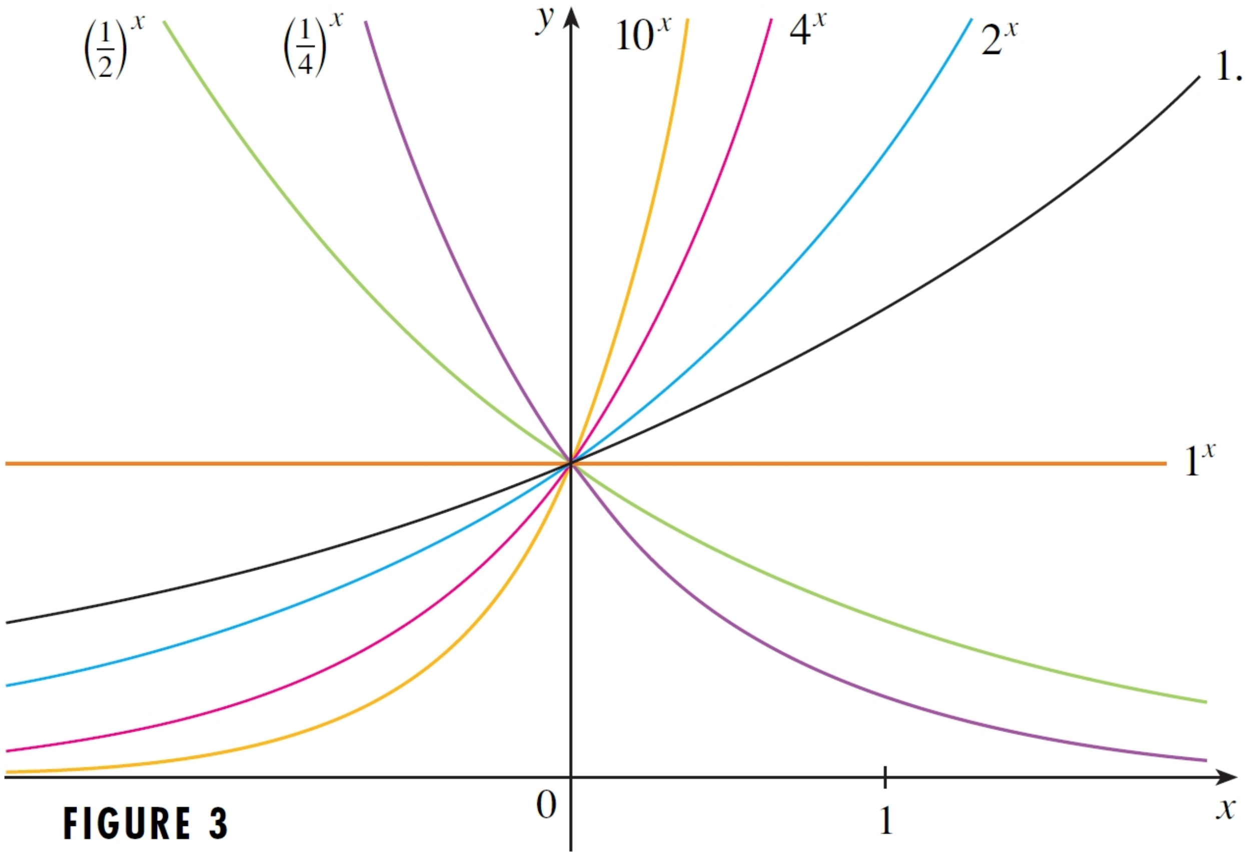


FIGURE 3

Laws of exponents

(1) $a^{m+n} = a^m a^n$

(2) $a^{m-n} = \dfrac{a^m}{a^n}$

(3) $(a^m)^n = (a^n)^m = a^{mn}$

(4) $(ab)^n = a^n b^n$

(5) $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$

(6) $\sqrt[n]{ab} = \sqrt[n]{a}\sqrt[n]{b}$

(7) $\sqrt[n]{\dfrac{a}{b}} = \dfrac{\sqrt[n]{a}}{\sqrt[n]{b}}$

(8) $(\sqrt[n]{a})^m = \sqrt[n]{a^m}$

(9) $(\sqrt[n]{a})^n = a$

Examples.

- $4^{\frac{1}{2}} = \sqrt[2]{4} = 2$
- $7^{-2} \cdot 7^{6} \cdot 7^{-4} = 7^{-2+6-4} = 7^{0} = 1$
- $10^{-3} = \frac{1}{10^3} = \frac{1}{1000}$
- $\frac{15^6}{15^5} = 15^{6-5} = 15^1 = 15$
- $(2^5)^2 = 2^{10} = 1024$
- $(3^{20})^{\frac{1}{10}} = 3^2 = 9$
- $8^{-\frac{2}{3}} = \dfrac{1}{(8)^{\frac{2}{3}}} = \dfrac{1}{(\sqrt[3]{8})^2} = \dfrac{1}{2^2} = \dfrac{1}{4}$

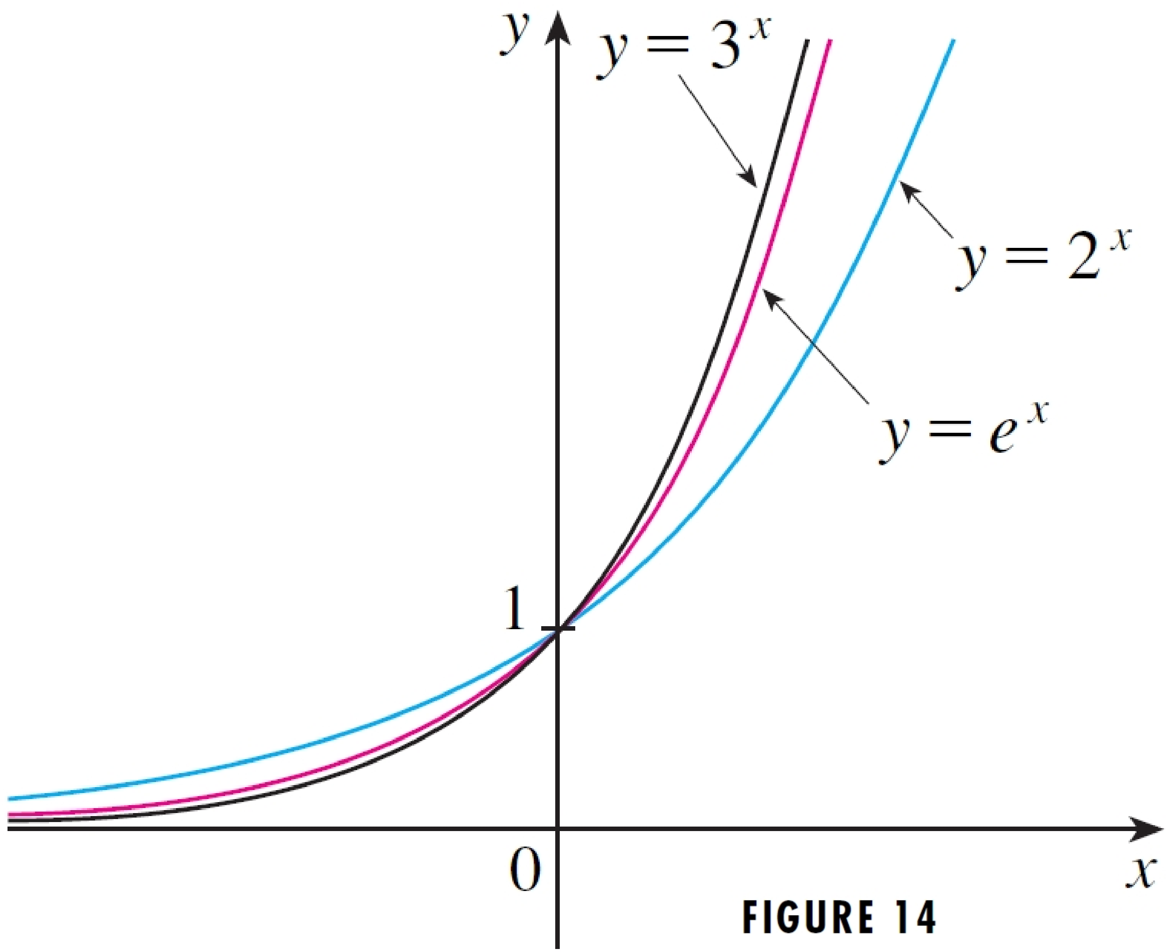


FIGURE 14

Section 1.5

Inverse functions and logarithms

(1) Inverse functions

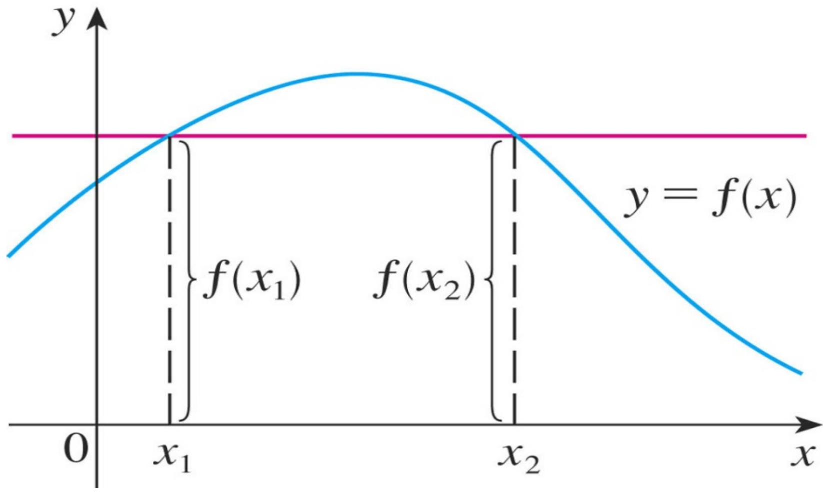
Definition: A function f is called a **one-to-one function** if it never takes on the same value twice; that is, $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$.

The Horizontal Line Test

A function is one-to-one if and only if no horizontal line intersects its graph more than once.

(1) Inverse functions (Cont')

***Example* 1.34**

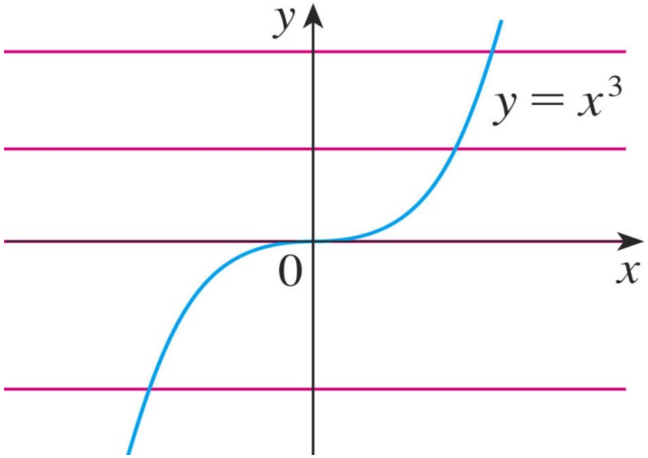


So, $f(x)$ is not a one-to-one function as $f(x_1) = f(x_2)$ and $x_1 \neq x_2$.

(1) Inverse functions (Cont')

***Example* 1.35**

Is the function $f(x) = x^3$ one-to-one?

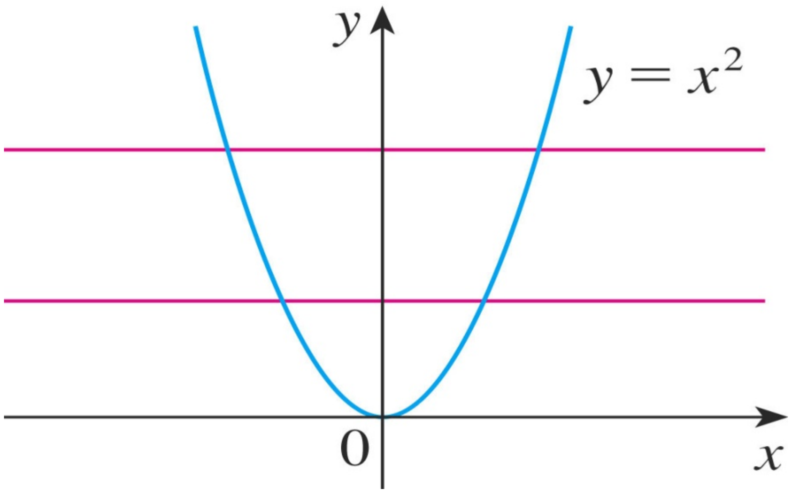


- Yes, because horizontal line intersects the graph of $f(x) = x^3$ only once.
- So, by the Horizontal Line Test, f is one-to-one.

(1) Inverse functions (Cont')

***Example* 1.36**

If the function $f(x) = x^2$ one-to-one?



- There are horizontal lines that intersect the graph of g more than once, e.g: $f(1) = 1 = f(-1)$,
- So, by the Horizontal Line Test, f is not one-to-one.

(1) Inverse functions (Cont')

Definition

Let f be a one-to-one function with domain A and range B. Then its **inverse function** f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \Leftrightarrow f(x) = y \text{ for any } y \text{ in B.}$$

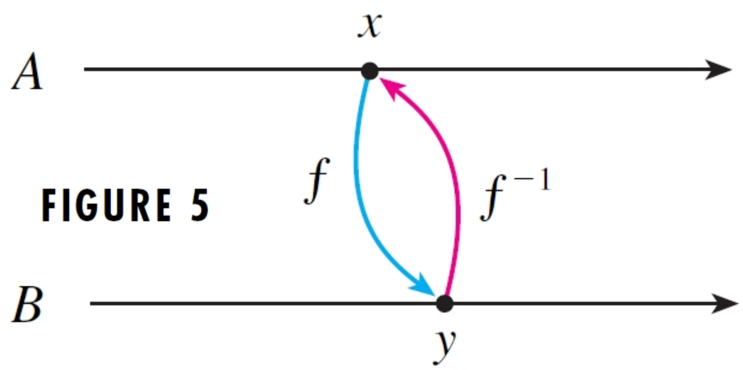


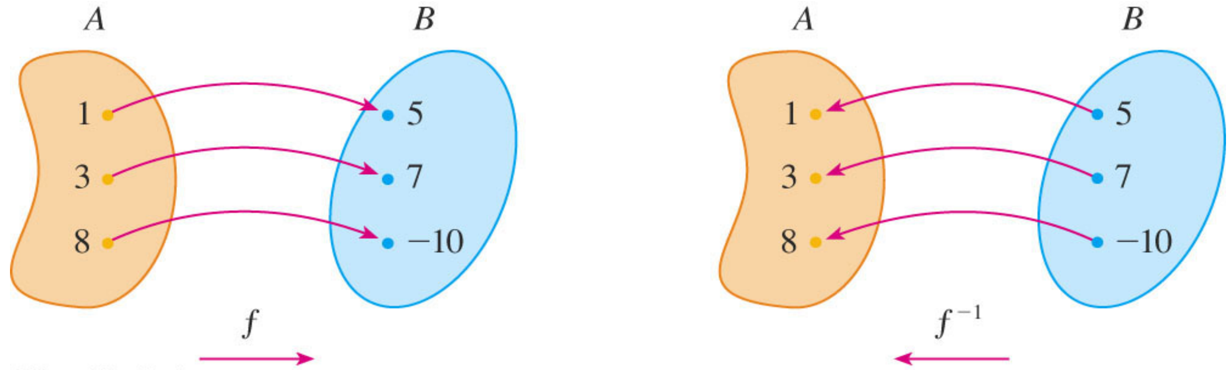
FIGURE 5

$$\left(f^{-1}(x) \neq \frac{1}{f(x)} \right)$$

(1) Inverse functions (Cont')

***Example* 1.37**

If $f(1) = 5$, $f(3) = 7$, and $f(8) = -10$, find $f^{-1}(7)$, $f^{-1}(5)$, and $f^{-1}(-10)$.



- From the definition of f^{-1}, we have:

$f^{-1}(7) = 3$ because $f(3) = 7$

$f^{-1}(5) = 1$ because $f(1) = 5$

$f^{-1}(-10) = 8$ because $f(8) = -10$

(1) Inverse functions (Cont')

- **<u>Properties of inverse function</u>**
- $f^{-1}(f(x)) = x$ for every x in the domain of f.
- $f(f^{-1}(x)) = x$ for every x in the domain of f^{-1}.

***<u>Example</u>* <u>1.38</u>**

Given $f(x) = x^3$, $f^{-1}(x) = \sqrt[3]{x}$, obtain $f^{-1}(f(x))$ and $f(f^{-1}(x))$.

$f^{-1}(f(x)) = (x^3)^{1/3} = x$

$f(f^{-1}(x)) = (x^{1/3})^3 = x$

(1) Inverse functions (Cont')

How to find the inverse function of a one-to-one function *f*

Step 1 Write $y = f(x)$.

Step 2 Solve this equation for x in terms of y (if possible)

Step 3 To express f^{-1} as a function of x, interchange x and y. Then $y = f^{-1}(x)$.

(1) Inverse functions (Cont')

- By definition of f^{-1}, a point (a, b) is on the graph of f if and only if the point (b, a) is on the graph of f^{-1}.

- Since we get the point (b, a) from the point (a, b) by reflecting about the line $y = x$, we find that

$$\boxed{\text{The graph of } f^{-1} \text{ is obtained by reflecting the graph of } f \text{ about the line } y = x.}$$

(1) Inverse functions (Cont')

***Example* 1.39**

Given $f(x) = x^3+2$. It is one-to-one function. Find $f^{-1}(x)$.

1)First, write $y = x^3 + 2$.

2)Then, we solve this equation for x : $x = \sqrt[3]{y-2}$

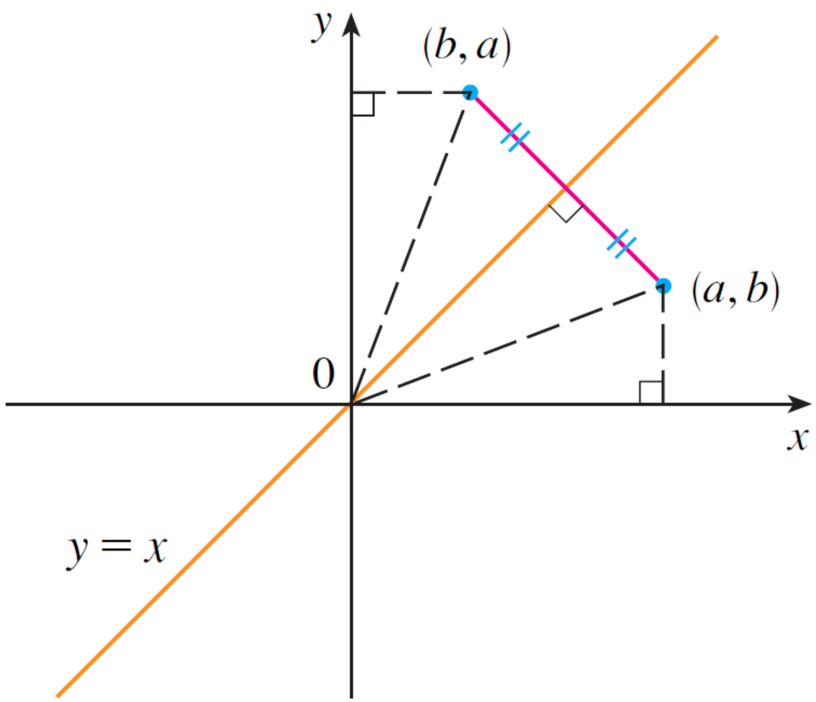
3)Finally, we interchange x and y : $y = \sqrt[3]{x-2}$

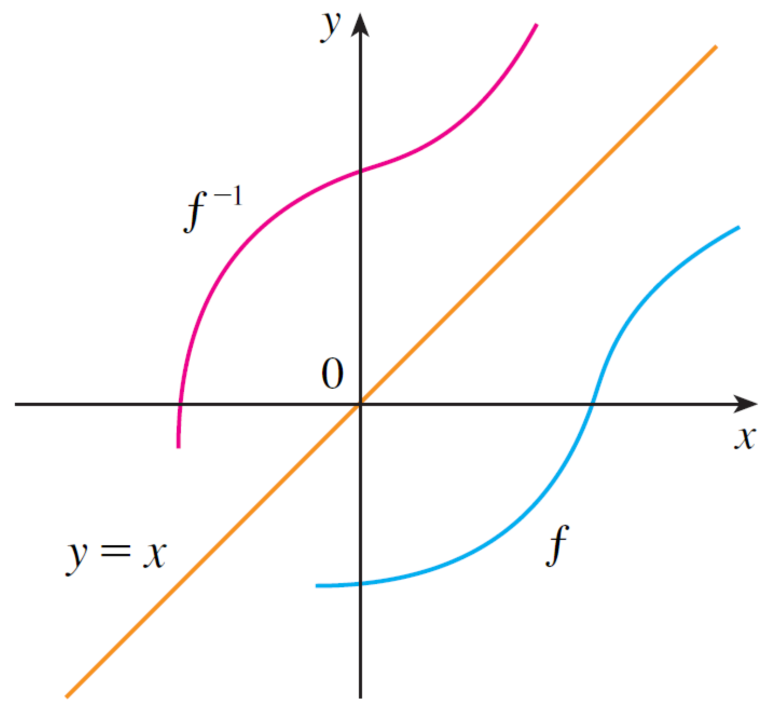
So, the inverse function is:

$$y = f^{-1}(x) = \sqrt[3]{x-2}$$

(1) Inverse functions (Cont')

Graph of f^{-1}:





$$y = f^{-1}(x) = \sqrt[3]{x \quad 2}$$

(2) Logarithmic functions

- If $a > 0$ and $a \neq 1$, the exponential function $f(x) = a^x$ is either increasing or decreasing and so it is one-to-one.
- Therefore it has an inverse function f^{-1}, which is called the logarithmic function with base a and is denoted by $\log_a$.
- $y = \log_a x \Leftrightarrow a^y = x$, for $a > 0$, $x > 0$

Properties of $y = \log_a x$

(1) $\log_a a^x = x$, for $x \in \Re$

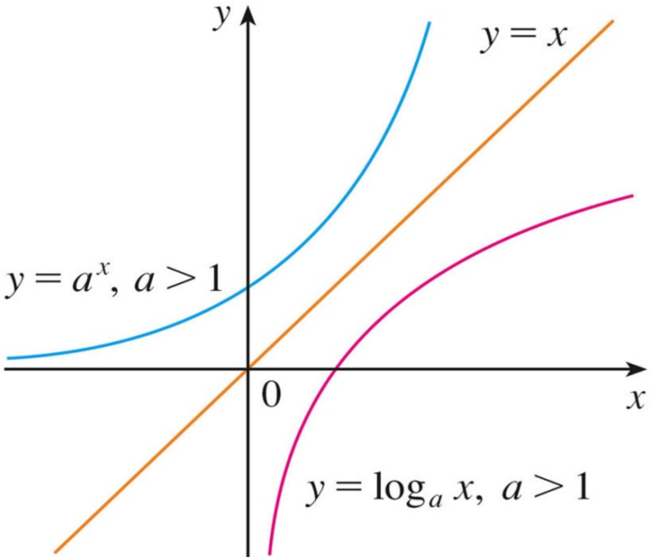
(2) $a^{\log_a x} = x$, for $x > 0$

(3) $\log_a 1 = 0$ $\quad a^0 = 1$

(4) $\log_a a = 1$ $\quad a^1 = a$

(2) Logarithmic functions (Con't)

Graph for $y = \log_a x$:



Note: $y = \log_a x$ is the inverse function for $y = a^x$. So the graph of $y = \log_a x$ is the reflection of the graph of $y = a^x$ about the line $y = x$.

Logarithm	
Exponential form	Logarithmic form
$b^x = y$	$\log_b y = x$
$2^4 = 16$	$\log_2 16 = 4$
$5^{-2} = 0.04$	$\log_5 0.04 = -2$

Logarithm = Exponent

$$\log_a N = x \longleftrightarrow N = a^x$$

(Common Log) $\log N = x \longleftrightarrow N = 10^x$

(Natural Log) $\ln N = x \longleftrightarrow N = e^x$

(2) Logarithmic functions (Con't)

<u>Laws of Logarithms</u>

(1) $\log_a bc = \log_a b + \log_a c$

(2) $\log_a \frac{b}{c} = \log_a b - \log_a c$

(3) $\log_a b^c = c \log_a b$

(4) $\log_a b = \frac{\log_c b}{\log_c a}$ (Change of base formula)

(5) $\log_a b = \frac{1}{\log_b a}$

- The logarithm with base 10 is denoted as $\lg x = \log_{10} x$
- The logarithm with base e is called the *natural logarithm* and is denoted as $\ln x = \log_e x$
- We may then define function such as $y = \ln x$, $\log_2 x$, etc.

(2) Logarithmic functions (Con't)

***Example* 1.40**

Use the properties of Logarithms to evaluate the following.

(a) $\log_4 2 + \log_4 32$

$$\log_4 2 + \log_4 32 = \log_4(2 \cdot 32)$$

$$= \log_4 64$$

$$= 3$$

(b) $\log_2 80 - \log_2 5$

$$\log_2 80 - \log_2 5 = \log_2\left(\frac{80}{5}\right)$$

$$= \log_2 16$$

$$= 4$$

(2) Logarithmic functions (Con't)

***Example* 1.40 (Cont')**

(c) Calculate $\log_5 89$. (Change of Base)

$$\log_5 89 = \frac{\log 89}{\log 5} \approx \frac{1.94939}{0.69897} = 2.7889$$

$$\log_5 89 = \frac{\ln 89}{\ln 5} \approx \frac{4.4886}{1.6094} = 2.7889$$

(d) Find the $\log_2 32$.

$$\begin{aligned} &\log_2 32 \\ &= \log_2 2^5 \\ &= 5 \log_2 2 \\ &= 5 \end{aligned}$$

(2) Logarithmic functions (Con't)

***Example* 1.40 (Cont')**

(e) Express $\log_{10}\left[\dfrac{\sqrt{10x}}{y^2}\right]$ in terms of $\log_{10} x$ and $\log_{10} y$.

$$\log_{10}\left[\frac{\sqrt{10x}}{y^2}\right]$$

$$= \log_{10}\sqrt{10x} - \log_{10} y^2$$

$$= \frac{1}{2}\log_{10}(10x) - 2\log_{10} y$$

$$= \frac{1}{2}\log_{10} 10 + \frac{1}{2}\log_{10} x - 2\log_{10} y$$

$$= \frac{1}{2} + \frac{1}{2}\log_{10} x - 2\log_{10} y$$

(2) Logarithmic functions (Con't)

Example **1.40 (Cont')**

(e) Express $\ln 20 - \ln 18 + 2\ln 6$ as a single logarithm.

$$\ln 20 - \ln 18 + 2\ln 6$$

$$= \ln 20 - \ln 18 + \ln 6^2$$

$$= \ln\left(\frac{20}{18} \times 36\right)$$

$$= \ln 40$$

(3) Natural logarithm

$y = \log_e x = \ln x$, for $x > 0$

Properties

(1) $\ln x = y \Leftrightarrow e^y = x$

(2) $\ln(e^x) = x$, for x

(3) $e^{\ln x} = x$, for $x > 0$

(4) $\ln e = 1$

(5) $a^x = e^{\ln(a^x)} = e^{x \ln a}$ for any base $a > 0$.

(3) Natural logarithm (Cont')

***Example* 1.41**

Find the solution to the equation $e^{5-3x} = 10$.

$$\ln(e^{5-3x}) = \ln 10$$

$$5 - 3x = \ln 10$$

$$3x = 5 - \ln 10$$

$$x = \frac{1}{3}(5 - \ln 10)$$

As the natural logarithm is found on scientific calculators, we can approximate the solution to four decimal places: $x \approx 0.8991$.

➢ For any positive number a ($a \neq 1$), we have $\log_a x = \dfrac{\ln x}{\ln a}$.

Section 1.6

Parametric curves

Parametric curves

Definition: A point (x, y) along the graph $y = f(x)$ can be described in a parameter t, which is called parametric equations, i.e. $x = f(t)$, $y = g(t)$.

- We can then interpret $(x, y) = (f(t), g(t))$ as the position of the particle at time t.

Parametric curves (Cont')

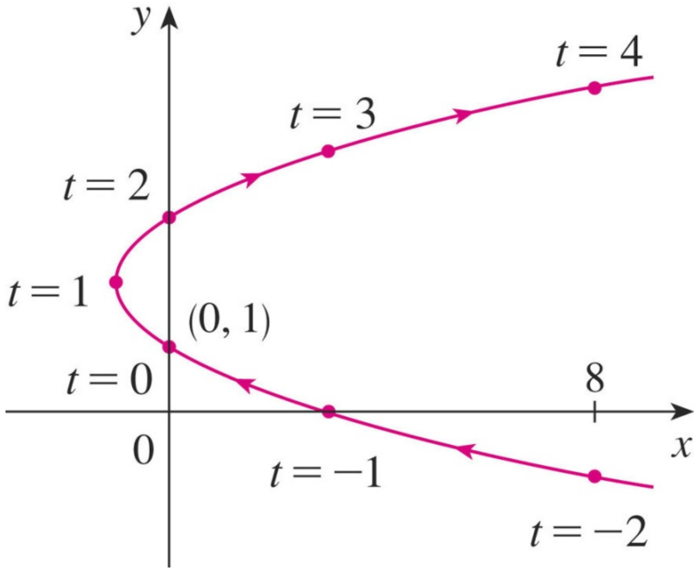
***Example* 1.42**

Sketch and identify the curve defined by parametric equations below:

$x = t^2 - 2t$

$y = t + 1, t \in \Re$

t	x	y
-2	8	-1
-1	3	0
0	0	1
1	-1	2
2	0	3
3	3	4
4	8	5



- This gives: $x = t^2 - 2t$

  $= (y - 1)^2 - 2(y - 1)$

  $= y^2 - 4y + 3$

- So, the curve represented by the given parametric equations is the parabola $x = y^2 - 4y + 3$.

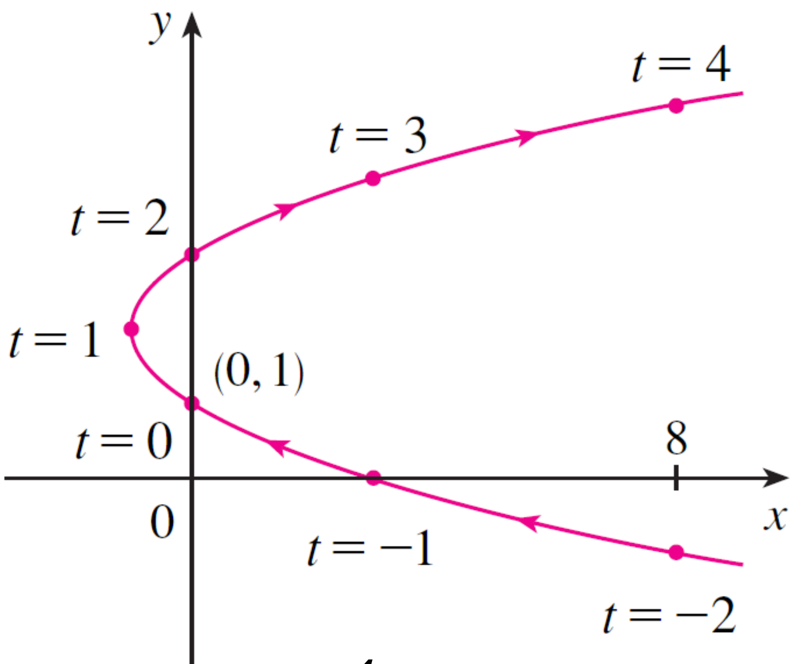
Parametric curves (Cont')

***Example* 1.42 (Cont')**

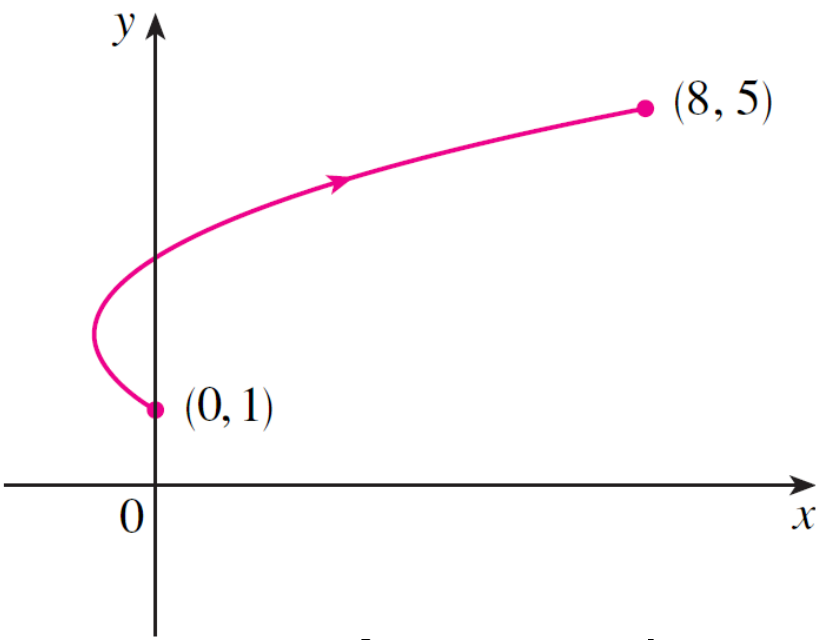
- Placed restrictions on the parameter t in the preceding example.
- For instance, the restriction $0 \le t \le 4$ would lead to just that portion of the parabola that starts at the point $(0, 1)$ and ends at the point $(8, 5)$.
- This is shown on the next slide:

Parametric curves (Cont')

Example 1.42 (Cont')



$t \in \Re$



$0 \leq t \leq 4$

Parametric curves (Cont')

***Example* 1.43**

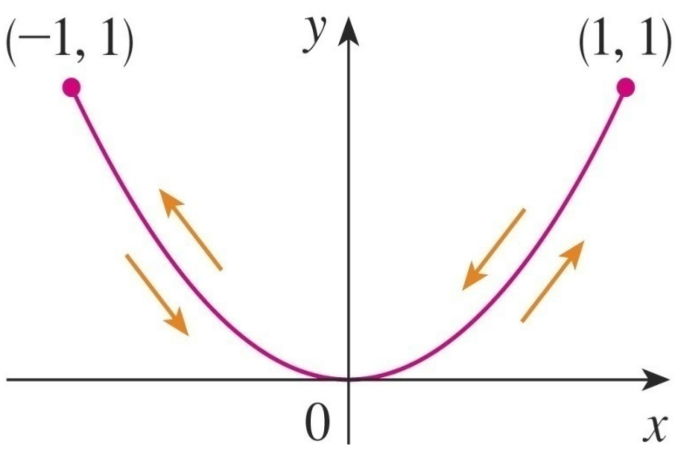
Sketch the curve $x = \sin t$, $y = \sin^2 t$

- So, $y = (\sin t)^2 = x^2$
- Thus, the point (x, y) moves on the parabola $y = x^2$
- As $-1 \leq \sin t \leq 1$, we have $-1 \leq x \leq 1$.
- So, the parametric equations represent only the part of the parabola for which $-1 \leq x \leq 1$.
- When $t = 0, x = 0, y = 0$

  $t = \pi/2, x = 1, y = 1$

  $t = \dfrac{3\pi}{2}, x = -1, y = 1$

  $t = 2\pi, x = 0, y = 0$



Parametric curves (Cont')

***Example* 1.44**

Sketch the curve $x = \cos t$, $y = \sin t$, and $0 \le t \le 2\pi$.

- $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$
- Thus, the point (x, y) moves on the unit circle $x^2 + y^2 = 1$

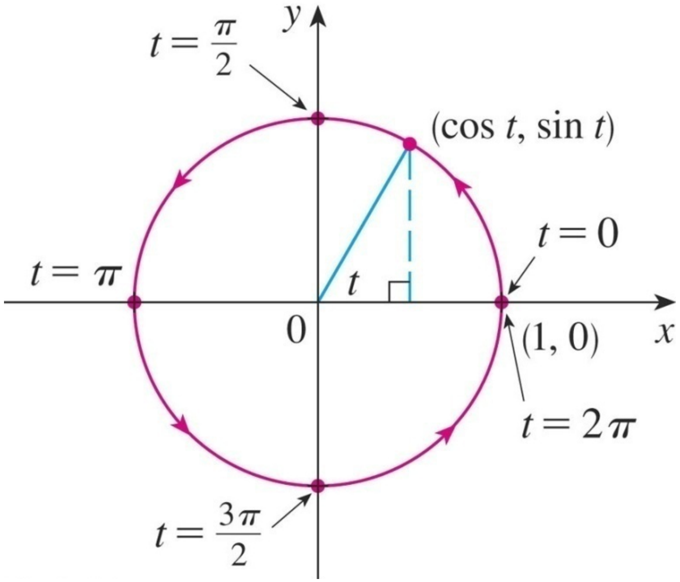
- When $t = 0, x = 1, y = 0$

  $t = \pi/2, x = 0, y = 1$

  $t = \pi, x = -1, y = 0$

  $t = \dfrac{3\pi}{2}, x = 0, y = -1$

  $t = 2\pi, x = 1, y = 0$



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Thank you